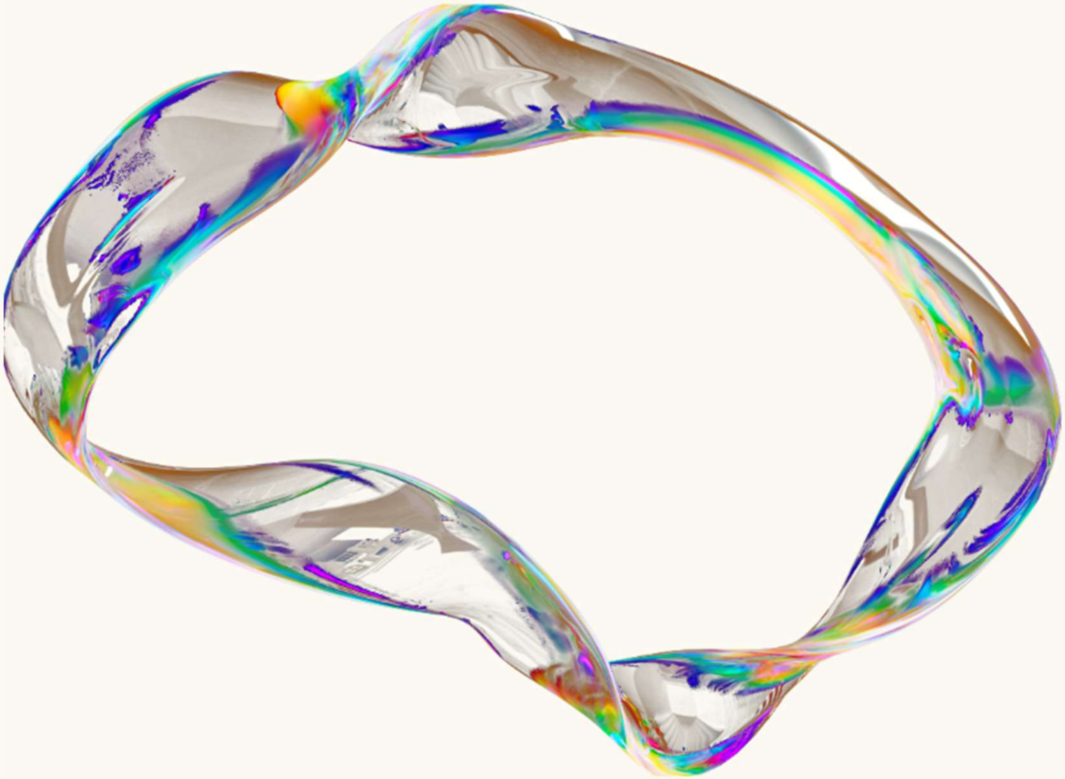

FROM CO_2 EMISSIONS TO OLEFINS

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1 Project Charter

Objective

To design a process that utilizes Carbon Dioxide (CO_2) and converts it to methanol through catalytic hydrogenation, raising methanol into light olefins through a Honeywell-UOP based MTO system modeling an economic, environmental, technologic performance of the proposed pathway.

Deliverables

- Block frow diagram and preliminary PFDs.
- Mass and energy balances Equipment descriptions and operating conditions.
- Environmental assessment.
- Technological/economical evaluation.
- Gantt chart.
- Final report and presentation.

Criteria for Successful Project

- Achieve 80,000 tons/yr olefins with 99.9% purity for industrial use.
- Evaluate 45Q credit impacts and CO_2 reduction
- Demonstrate the design of an efficient alterative to create ethylene propylene to replace fossil-fuel based production
- Explore feasible alternative processes to Honeywell-UOP MTO process for future development of the process.

Project Scope

- CO_2 repurposes to reduce carbon emissions.
- Integration of Hydrogen (SMR + capture/external supply).
- Methanol synthesis over ZnO/Cu/Al2O3,
- MTO processing with SAPO-34 method.
- Olefin purification to polymer grade.
- Alternative pathways from Olefin production.
- Market, environmental, safety, and economic analysis.

Executive Summary

The global chemical industry is undergoing a major transition driven by increasing demand for essential materials and growing pressure to reduce carbon emissions. Light olefins, particularly ethylene and propylene, are among the most important building blocks in modern manufacturing, supporting industries such as plastics, packaging, construction, automotive, and consumer goods. Demand for these materials continues to rise steadily at approximately 3–4% annually, and long-term projections indicate sustained growth over the next decade. However, conventional production methods rely heavily on fossil fuels and generate significant greenhouse gas emissions. This creates both a challenge and an opportunity: the need to develop alternative production pathways that are both economically competitive and environmentally sustainable.

This project presents a fully integrated process that transforms captured carbon dioxide (CO_2) into high-value olefin products, offering a practical pathway toward a circular carbon economy. Rather than treating CO_2 as a waste product, the proposed design repositions it as a valuable feedstock. By combining carbon capture, hydrogen production, methanol synthesis, and methanol-to-olefins (MTO) conversion within a single facility, the process creates a closed-loop system that reduces emissions while generating commercially viable products. This approach aligns closely with global sustainability goals and positions the company as a leader in next-generation chemical manufacturing technologies.

The proposed facility is strategically located in the Houston, Texas region, one of the largest petrochemical hubs in the world. This location provides several key advantages, including direct access to established CO_2 pipeline infrastructure, reliable hydrogen supply networks, and abundant natural gas resources. Additionally, proximity to major industrial emitters allows for efficient CO_2 capture from flue gas streams, reducing feedstock costs and ensuring a consistent supply of raw materials. The region also benefits from a highly developed logistics network, skilled workforce, and supportive regulatory environment, all of which contribute to reduced operational risk and improved project feasibility.

The plant is designed to produce approximately 80,000 tons per year of polymer-grade olefins, primarily ethylene and propylene, with purities exceeding 99.9%. These products are essential inputs to produce polyethylene, polypropylene, and a wide range of downstream materials. In addition to the primary products, the process also generates byproducts such as C_4 hydrocarbons, which can be sold or further processed, providing additional revenue streams. Market analysis indicates that both the ethylene and propylene markets are expanding significantly, with increased emphasis on locally produced, low-carbon alternatives. As companies and consumers place greater importance on sustainability, there is a growing premium for materials produced with reduced environmental impact, further strengthening the competitive position of this process.

From a technical perspective, the process integrates several well-established industrial technologies into a cohesive and highly efficient system. Hydrogen is produced via steam methane reforming (SMR) and purified to meet stringent specifications. Carbon dioxide is captured from process streams and combined with hydrogen in a catalytic reactor to produce methanol, an intermediate chemical that serves as a bridge between carbon capture and olefin production. Methanol is then converted to light olefins in a fluidized-bed MTO reactor, enabling high conversion efficiency and continuous catalyst regeneration. Downstream purification steps, including CO_2 removal, molecular sieve drying, membrane separation, and cryogenic distillation, ensure that final products meet polymer-grade quality standards. A key feature of the design is the recycling of CO_2 within the process, which reduces waste emissions and improves overall process efficiency.

The economic evaluation demonstrates that the project is both viable and financially attractive. The total capital investment is estimated at approximately $740 million, reflecting the scale and complexity of the integrated system. While hydrogen production and CO_2 capture account for the largest share of capital costs due to their energy-intensive nature, these investments are essential to enabling the circular carbon pathway. Despite the high initial investment, the plant generates strong financial returns, with annual revenue estimated at approximately $424 million. A significant portion of this revenue is derived from the sale of propylene and ethylene, while additional contributions come from byproducts and government incentives. Notably, the inclusion of 45Q carbon capture tax credits provides over $120 million in annual tax credits, significantly enhancing the project's economic performance.

The process achieves a gross profit of over $300 million per year, demonstrating strong profitability potential. Net present value (NPV) analysis indicates that the project reaches breakeven and becomes profitable after approximately 12 years, which is consistent with large-scale chemical plant investments. Beyond this point, the project generates steadily increasing cash flows, delivering long-term value to stakeholders. Further sensitivity analysis shows that product pricing and revenue have the greatest impact on profitability, highlighting the importance of market conditions and pricing strategies. However, even under conservative assumptions, the project remains economically robust, particularly when supported by carbon capture incentives.

In addition to its financial performance, the project offers significant environmental and strategic benefits. By capturing and reusing CO_2, the process substantially reduces net greenhouse gas emissions compared to conventional olefin production methods. This positions the company to meet current and future regulatory requirements, including emissions-reduction targets and carbon-pricing mechanisms. Furthermore, the ability to produce low-carbon olefins provides a competitive advantage in markets increasingly driven by sustainability considerations and ESG commitments. The integration of carbon capture and utilization technologies also enhances the company’s reputation as an innovator in sustainable chemical production.

Based on the comprehensive analysis conducted in this study, it is recommended that the company proceed with further development and potential commercialization of the CO_2-to-olefins process. Attention should be given to optimizing hydrogen production and carbon capture efficiency, as these represent the most significant cost drivers. Additionally, exploring opportunities for downstream integration, such as polymer production, could further increase profitability and reduce exposure to market volatility. Overall, this project represents a strategically compelling investment opportunity, combining strong market demand, favorable policy incentives, advanced process technology, and meaningful environmental impact.

2 Introduction

The global demand for light olefins, particularly ethylene and propylene, continues to grow due to their essential role as building blocks in polymers, chemicals, and consumer materials. However, conventional monomer production using fossil fuels contributes significantly to global CO_2 emissions and environmental degradation. As the chemical industry remains a major source of industrial greenhouse gases, there is an urgent need for more sustainable production alternatives.

To address the challenge, this project explores an alternative approach that integrates carbon capture with chemical conversion technologies to transform CO_2 emissions into valuable olefin monomer products. By utilizing captured carbon dioxide as a feedstock, combined with hydrogenation and methanol-to-olefins (MTO) processes, the system creates a circular carbon pathway that reduces reliance on fossil resources while lowering overall emissions.

This approach not only mitigates environmental impact but also presents economic opportunities by converting waste CO_2 into marketable chemicals and polymers. Supported by favorable policy incentives and growing demand for sustainable materials, CO_2 derived olefin production represents a promising pathway toward a low carbon, circular chemical industry.

2.1 Problem and Significance

Global demand for light olefins, primarily ethylene and propylene, has steadily increased due to their crucial role as precursor building blocks for industries such as polymers, fibers, chemicals, and consumer materials. The International Energy Agency (IEA) reports that olefin consumption continues to rise at 3% to 4% annually, (Commodity Plastics Market Size to Worth USD 666.76 Billion by 2034 2025) making petrochemicals one of the fastest-growing sources of oil demand worldwide. Traditionally, ethylene and propylene are produced through steam cracking of hydrocarbons derived from crude oil and natural gas. This conventional method is highly energy-intensive, environmentally damaging, and responsible for substantial CO_2 emissions and industrial waste. For example, ethylene production alone releases approximately 1.5–3.0 tons of CO_2 per ton of product, depending on feedstock and energy source.

These fossil-based processes contribute heavily to global greenhouse gas emissions, resource depletion, and worsening climate impacts. The chemical industry now accounts for 18% of global industrial CO_2 emissions, with light olefin production being a major contributor. (Polymers Market Strengthen Industrial Application with Bio-Based Materials 2025)

To address this issue, there is a growing focus on closing the carbon loop. Rather than releasing CO_2 during olefin production, captured CO_2 can be used as a carbon feedstock, providing a green alternative to petrochemical extraction and mitigating emissions from existing industrial activity.

2.1.1 Carbon Capture Implementation

Carbon Capture and Repurposing is recognized as a critical climate mitigation pathway by global scientific and policy organizations. Industrial flue gas, especially from energy-intensive manufacturing such as oil refining, contains large volumes of CO_2 at concentrations that make flue gas capture economically viable. CO_2 emissions from Oil Refining alone contribute 13 to 15% of global CO_2 emissions. (Fagelson 2024) Within the United States, Oil Refining is one of the largest industries that can be used to capture CO_2. Such as in Galveston Bay, Texas, which is home to one of the largest integrated oil refineries in the world, producing millions of tons of flue-gas CO_2 annually. This provides a convenient, continuous source of carbon for CO_2 capture to produce valuable chemical products. In 2024, MGBR produced 8.2 million tons of GHG. This accounted for approximately 0.5% of global GHG emissions. Reducing and using this source to create polymers can contribute to a more sustainable world**.** (Fagelson 2024)

By integrating carbon capture with chemical conversion technologies such as CO_2 hydrogenation and methanol-to-olefins (MTO) synthesis, previously emitted carbon can now be preserved within a circular process. This reduces direct emissions from oil refinery operations, dependence on fossil hydrocarbons, and net global atmospheric carbon intensity. The approach not only prevents waste emissions but transforms CO_2 into a revenue-generating feedstock, aligning environmental sustainability with industrial profitability for a viable plant design and implementation.

2.1.2 Sustainable Olefin Production

Sustainable olefin manufacturing is a major transformative objective across the chemical and consumer product industries. Traditional plastics rely on fossil fuels, which generate massive amounts of waste. By contrast having a sustainable olefin manufacturing process can prioritize the use of Carbon emissions, advanced catalysis to lower energy consumption and improve selectivity, circular-economy recycling methods, and low emission energy integration across chemical plants Organizations such as the National Academies and IEA note that integrating CO_2 utilization into chemical value chains could reduce global carbon emissions by billions of tons annually while supporting economic growth. (Polymers Market Strengthen Industrial Application with Bio-Based Materials 2025)

Sustainable polymer production, lowers GHG emissions across the entire product lifecycle, reduces reliance on crude-oil-derived hydrocarbons, encourages innovation in lightweight, high-performance materials, and supports global decarbonization legislation and ESG goals By converting CO_2 into monomers such as ethylene and propylene, materials already fully compatible with existing polymer infrastructure, this approach accelerates the transition toward a carbon-circular plastics economy without requiring redesign of consumer products.

2.2 Value Chain

The value chain for this project spans the transformation of captured carbon dioxide into a high-value polymer. This pathway leverages clean energy, catalytic conversion technologies, and established downstream polymer markets to create both economic and environmental value. The proposed plan begins with CO_2 as a feedstock, proceeds through methanol synthesis, and then to a methanol-to-olefin (MTO) conversion, ultimately producing light olefin monomers. At each stage, value is added, but costs and carbon-intensity risks also accumulate as the process. More importantly, the ultimate competitiveness and sustainability of the chain depend on two key drivers: the cost and carbon footprint of the hydrogen used, as well as the availability of policy incentives.

CO_2 from stored pipelines is fed into a hydrogenation loop where H_2 and CO_2 react to produce methanol. The hydrogen supplied is generated on-site through SMR with carbon capture. This methanol serves as an intermediate liquid that will substitute conventional fossil-derived methanol. In methanol synthesis, the core MTO reactor converts methanol into light olefins (ethylene and propylene) and generates byproducts. These olefins serve as the foundational building blocks for polyolefin elastomers, solvents, and numerous petrochemical derivatives.

The process converts waste CO_2 to high-value commodity chemicals, effectively recycling CO_2 emissions. The marketing of this product will focus on CO_2 recycling and on providing a lower-carbon footprint for environmentally friendly materials. With Certification, emissions disclosure, and compliance with circular-economy targets, create strategic market differentiators within the olefin market. The incentivized to adopt these materials to meet corporate ESG goals and comply with regulatory mandates regarding the use of recycled or low-carbon plastics.

However, the value of this chain is susceptible to complications from the hydrogen source. Since hydrogen is produced by conventional steam-methane reforming, without effectively producing grey hydrogen. In that case, upstream CO_2 reductions may be offset by CO_2 emitted during hydrogen production, undermining any climate advantage and increasing the net carbon footprint. To reduce the plant's carbon footprint, hydrogen must be produced on-site while simultaneously capturing the carbon and using it as a supplemental feedstock for olefin production.

The implementation of carbon capture in SMR hydrogen production enables the use of federal carbon capture tax credits. The economics of this chain are also heavily dependent on the government and regulatory incentives around carbon capture and utilization. In a U.S. context, the tax credit framework under Section 45Q offers financial incentives per metric ton of CO_2 captured and either stored or utilized (Jones 2023) Under Section 45Q, a facility that captures CO_2 and uses it in chemical conversion may qualify for the utilization credit rate (Jones 2023)Section 45Q tax credits provide up to \$60 per metric ton of CO_2 utilized and up to \$130 per metric ton stored permanently, depending on the pathway and compliance requirements, according to congressional analysis. This incentive may significantly improve the economics of upstream carbon capture, provided the facility meets the eligibility criteria. A crucial consideration in this value chain is that monomers alone do not maximize profitability.

Ethylene and propylene are globally traded commodities with volatile margins and are most economically valuable when converted into higher-value derivatives, such as polymers. Polyethylene and polypropylene command significantly higher selling prices and more stable demand profiles than their monomer precursors; therefore, a strategic integration of downstream polymerization units is necessary to fully capture the value of this conversion process. Without access to polymerization, the project would remain exposed to monomer price cyclicality and face reduced returns on investment. Given these dependencies, the value of this project hinges on hydrogen sourcing costs, carbon intensity, final product form, and the policy environment.

With localized hydrogen production and a favorable 45Q carbon capture tax credit, CO_2 derived olefins can become economically competitive and environmentally attractive. Without government tax credits that incentivize carbon capture, the process will not be able to compete with conventional, low-cost fossil-fuel olefin production. This value chain demonstrates how the transformation of low-value CO_2 into polymer materials represents a complete industrial upgrade pathway.

2.3 Raw Materials and Utilities Characteristics

The selection, cost, and sourcing strategies for raw materials are critical drivers of both the economic and environmental performance of the proposed CO_2 to polymers facility. The major inputs to this process include Carbon Dioxide and Hydrogen from the pipeline feedstock, Natural Gas, Water, and Electricity as Utilities, and CuO-ZnO-Al2O3, MDEA, and SAPO-34 as Catalysts for operating the CO_2 to Olefin plant.

2.3.1 Feedstocks

The primary feedstocks for the process are carbon dioxide and hydrogen, both sourced locally in the Houston area via established industrial pipeline networks. CO_2, priced at \$760 per ton and supplied via the Kinder Morgan pipeline system (Kinder Morgan n.d.), serves as the carbon source for the olefin production. Hydrogen, at \$3740 per ton and delivered through the Air Liquide pipeline network (Air Liquide n.d.), is used to convert CO_2 into Methanol for Olefin production.

2.3.2 Catalysts

Catalysts for the CO_2-to-olefin process are sourced from China and are essential at different stages of the production process. MDEA, priced at $1034 per ton (Deshang Chemical n.d.), is used as a solvent in gas treatment systems, where it selectively absorbs CO_2 from gas streams for carbon capture. SAPO-34, priced at $34 per 5 tons (ZR Catalyst n.d.), will function as a catalyst in Methanol to Olefins (MTO) processes, where it will convert methanol into light olefins, ethylene, and propylene. $CuO/ZnO/Al_2O_3$ catalyst, priced at $132 per barrel (HNYT Carbon n.d.), is used in methanol synthesis to enable efficient hydrogenation reactions and improve overall conversion efficiency.

2.3.3 Utilities

Utility costs for the Houston area for operating the CO_2 to Olefin production plant are as follows: Natural gas, priced at $4.67 per cubic foot (Energy Information Administration n.d.), serves as the primary heating utility, supplying the thermal energy required for reactors, distillation columns, and other high-temperature operations. Electricity at $0.07 per kW powers (Power Wizard n.d.) machinery such as pumps, compressors, and control systems, making it essential for maintaining continuous plant operation and process control. Water, costing $2.50 per 5,000 gallons (Harris County MUD 208 n.d.), is a critical utility for cooling systems and steam generation, which require heat exchange and energy integration.

Table 1: Pricing Materials and Supplies

Product	Market Value: USD	Market Information	Use in process
MDEA	1034 per ton	Source from China	Catalyst for CO_2 extraction.
SAPO-34	34 per 5 tons	Source from China	Catalyst for Methanol production.
CuO-ZnO-Al2O3	132 per barrel	Source from China	Catalyst for Olefin production.
CO_2	760 per ton	Houston Kinder - Morgan pipeline	Feedstock.
H_2	3740 per ton	Houston Air Liquide pipeline	Feedstock.
Natural Gas	4.67 per cubic foot	Houston area price	Utility for heating.
Electricity	0.07 per KW	Houston area price	Utility for powering machinery.
Water	2.50 per 5000 gallons used	Houston area price	Utility for cooling and steam generation.

2.4 Product Characteristics

With the capture and processing of CO_2 into valuable products, the design basis intends to create an operable plant that produces industrial-grade olefin monomers. With the expanding market for polymer development and the scarcity of petrochemical products due to recent geopolitical events, the need to develop alternative sources of monomers and petrochemical products must be addressed to meet demand and alleviate the shortage. With vast amounts of CO_2 stored in the surrounding Houston area, it can be used to meet the plant's requirements for producing olefin monomers.

The economic viability of operating a Carbon Capture to MTO-based plant depends on the market value of sellable products. This process produces two olefin monomers, ethylene and propylene, as well as n byproducts that can be sold for profit.

As of late 2025, the market value of Polymer Grade Ethylene Monomer in the North American exchange region is USD 0.52 per kilogram (Intratec n.d.) . Polymer Grade Propylene Monomer market prices are assessed at USD 1.27 per kilogram (Intratec n.d.). The margins of profit for the sales of monomers are increased by the sales of byproducts produced through the process. The production of industry grade Butene is valued at approximately 1.05 USD per kilogram. (Intratec n.d.) The future profitability of product sales increases with the volatility of current geopolitical events. Therefore, the market value of petrochemical products is expected to rise due to a shortage.

Table 2: Pricing of Products

Product	Market Value: USD/ton	Market Information
Ethylene Monomer	520	North American Market Exchange
Propylene Monomer	1270	North American Market Exchange
Butene Gas	1050	North American Market Exchange

2.5 Regulatory Environment

Developing the CO_2-to-Olefins process must comply with environmental regulations focused on controlling emissions, ensuring safe handling of chemicals, and preventing environmental contamination. Air pollutants such as carbon dioxide, carbon monoxide, methane, and volatile organic compounds (“ethylene, propylene, butene, and pentane”) must be regulated under the Clean Air Act, which requires emission limits, monitoring, and permits. Large greenhouse gas emitters must also report emissions under the EPA Greenhouse Gas Reporting Program. Liquid emission regulations fall under the Clean Water Act, which requires wastewater containing substances like methanol or MDEA to be treated and discharged under NPDES permits to meet pollutant limits.

In addition, safety and waste management regulations play a key role. Hazardous chemicals like hydrogen are governed by OSHA Process Safety Management standards to prevent accidents. Solid materials such as catalysts "SAPO-34, MDEA, and $Cu/ZnO/Al_2O_3$: must be properly handled, recycled, or disposed of under the Resource Conservation and Recovery Act (RCRA) to avoid soil and groundwater contamination. Overall, regulations emphasize emission reduction, proper treatment of waste streams, safe handling of chemicals, and strict reporting requirements to ensure environmental and human health protection.

2.6 Supply, Demand, and Competitors

The global olefin industry is a crucial market that encompasses large parts of trade and industry due to the heavy reliance on plastics used in packaging, construction, automotive, textile, and consumer goods. The production of monomers relies heavily on fossil fuels, leading to prices that track oil and natural gas prices, which can fluctuate due to geopolitical factors. With a turbulent international scene impacting the global market for fossil fuels needed to produce monomers, the need for alternative monomer production processes and regionalization of monomer production is on the rise to meet demand. (Eco-Business n.d.)

A commonly used polymer, ethylene, has a market value of USD 198.2 billion in 2024. The ethylene industry is forecast to grow from USD 208.71 billion in 2025 to USD 349.87 billion by 2035, underscoring demand for expanding ethylene production over the upcoming decade. (Intratec n.d.)

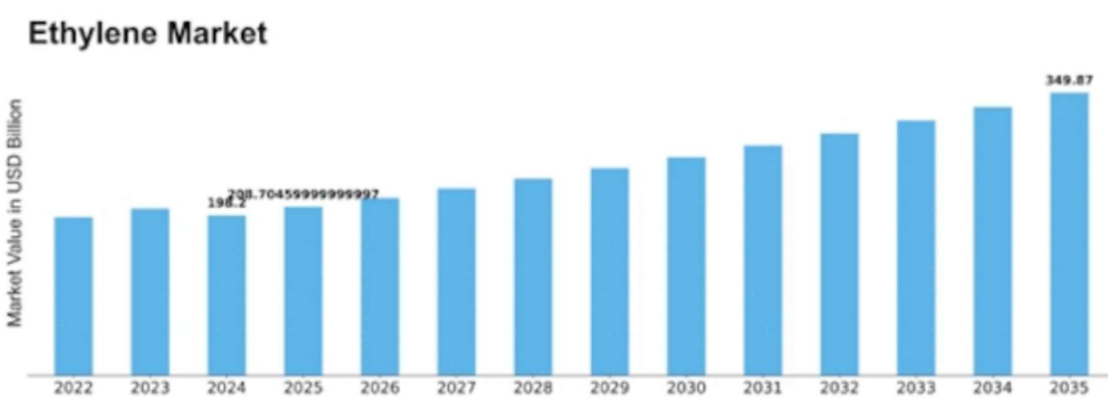

Figure 1. Ethylene Market Trend 2022-2035), (Intratec n.d.).

Currently, the Ethylene market is transitioning toward sustainability and technological innovation focused on greener, localized production. Having led to increasing pressure to reduce carbon emissions, it is driving interest in carbon-emission-free ethylene and regional economic practices such as recycling and CO2 recovery. At the same time, geopolitical factors and supply chain disruptions introduce uncertainty, particularly regarding the availability and pricing of fossil fuel feedstocks. This allows for future growth to depend heavily on how effectively it produces cleaner technologies and adapts to regulatory and environmental constraints. Overall, the ethylene market is expanding, with a focus on greener, regional production. (The Business Research Company n.d.)

Propylene is another commonly used monomer produced from fossil fuels that can also be produced locally. The propylene market's growth in North America reflects expansion driven by structural regional demand rather than reliance on international markets. The North American market is valued at $159 billion in 2026. The propylene market is expected to reach $198 billion by 2032, with an annual growth rate of approximately 5% to 6%. (Statsnex Market Insights n.d.) Over the decade, growth has been characterized by incremental volume increases rather than price-driven surges from fossil fuel and geopolitical influence, allowing the industry to expand local development and alternative methods in propylene production rather than international market changes from fossil fuel prices.

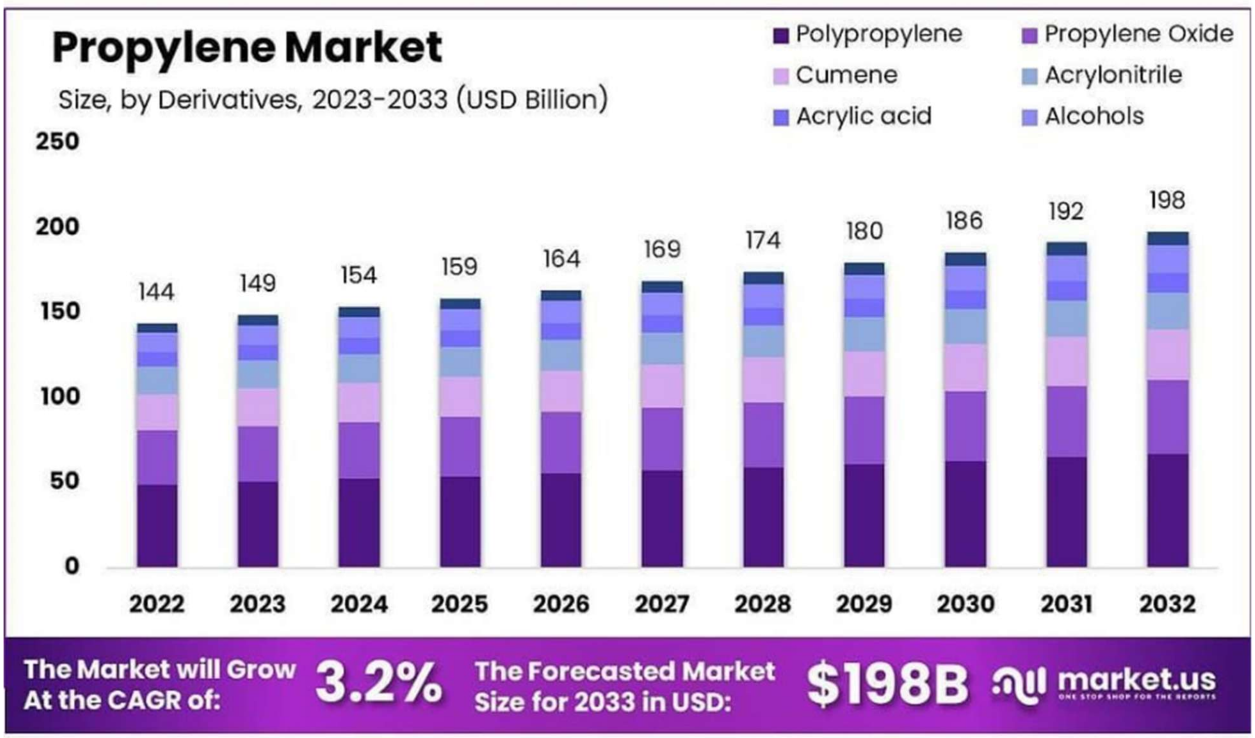

Figure 2. Propylene Market Outlook (2022-2032), (The Business Research Company n.d.) .

The North American propylene market's growth is focused on regional concentrated development and green carbon-free technology. This allows for the creation of green propylene plants centered in the US. The market in North America also remains competitive due to advantaged local feedstocks, with global conflicts forcing localization of propylene production the focus in expansion is increasingly shaped in building new infrastructure focused on greener production, to replace traditional fossil fuel based propylene production, As a result, the decade-long growth of the propylene market is not only focused on quantitative production output but also in creating local green alternatives to fossil fuel production of propylene.

The push toward local and greener propylene production technologies is justified by both economic resilience and environmental necessity. Developing regional production capacity reduces dependence on volatile global supply chains and feedstock imports for supply security over the long term. At the same time, greener technologies that involve carbon capture integration to lower emissions in the monomer production sector. Together, these initiatives support sustainable decade-scale market growth by aligning industrial expansion with environmental targets, while also positioning producers to remain competitive in a future defined by stricter carbon constraints and localized manufacturing priorities.

3 Judging Criteria

Due to rising global CO_2 emissions, up 0.8% in 2024 to an all-time high of 37.8 Gt CO_2, there is a strong incentive to develop alternative, more sustainable industrial processes. Much of this increase stems from industrial activity, which has contributed to a 50% rise in atmospheric CO_2 concentrations (ppm) since 1750**.** (Yusuf and Almomani 2023)

Currently, the combustion of natural gas, liquid fuels, and coal remains the primary driver of long-term carbon emission growth. Countries such as China, Japan, Indonesia, India, and the United States are among the largest contributors due to their substantial energy consumption. However, it is worth noting that the United States and Japan have seen a steady decline in CO_2 emissions since the early 2000s. (Polymers Market Strengthen Industrial Application with Bio-Based Materials 2025)

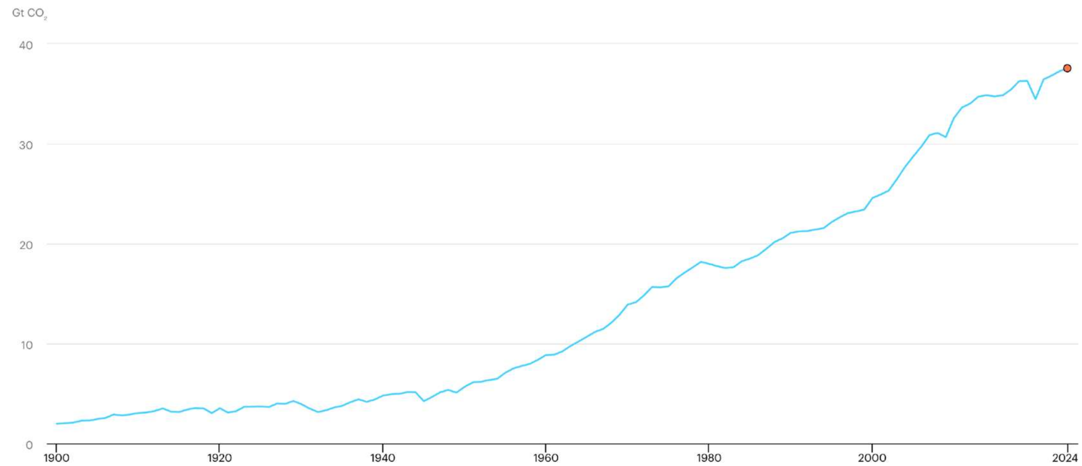

Figure 2. Global CO_2 Emissions Trend (1900-2024), (IEA 2025).

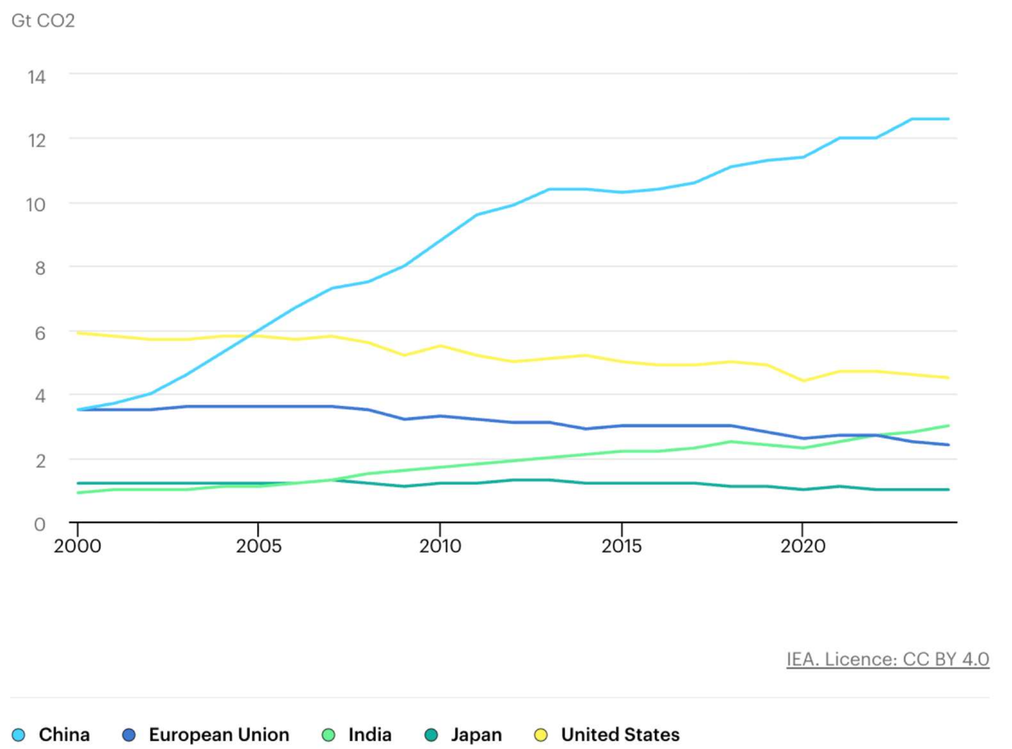

Figure 3. CO_2 Total Emissions by Region (2000-2024), (IEA 2025).

For our Methanol-to-Olefins (MTO) project, we considered the significant environmental impacts of carbon emissions. Our process begins by capturing CO_2 and converting it to methanol (MeOH) through hydrogenation. Rather than relying on conventional processes that generate additional flue gas from the combustion of natural gas or coal, our design utilizes burn stack flue gas from oil refinery methane combustion, from which CO_2 is purified and recovered.

Through heat integration and process optimization, the CO_2 emissions associated with MeOH production are reduced from 3.15×10^{-1} $tCO_2/tMeOH$ to 6.77×10^{-3} $tCO_2/tMeOH$, representing roughly a 98% decrease (see figure below). This reduction is supported by high process selectivity, including a CO_2 conversion of approximately 99%, enabled by an optimized CO_2:H_2 ratio (1:7) and an industrially relevant hydrogenation catalyst. (Yusuf and Almomani 2023)

Downstream, the produced MeOH is converted into light olefins using a proprietary SAPO-34 catalyst. During this reaction, the catalyst gradually accumulates solid carbon ("coke"). To maintain catalyst performance, SAPO-34 undergoes regeneration by controlled combustion, which produces additional CO_2. In our design, this CO_2 is captured and recycled back into the MeOH synthesis loop, further lowering net emissions while also providing a modest increase in overall plant profitability.

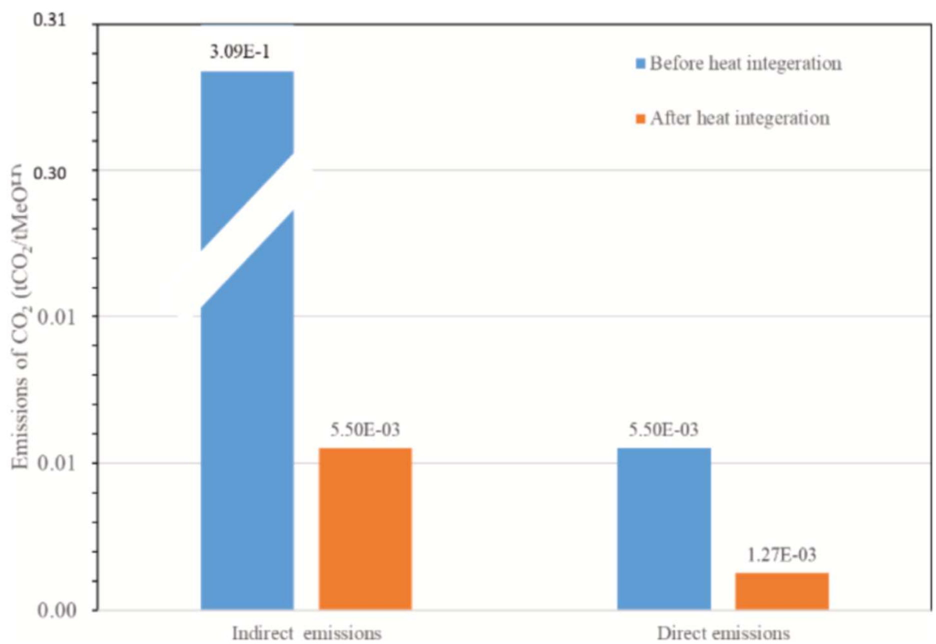

Figure 4. Total Estimated CO_2 Emissions for the Hydrogenation Process Before and After Heat Integration (Yusuf and Almomani 2023).

Our project also prioritizes efficient energy use. As mentioned earlier, the initial step of our process involves capturing CO_2 from steel-mill off-gases using an amine-based absorbent solution. Conventional amine absorption is highly energy-intensive because CO_2 absorption is exothermic, while regeneration is endothermic, requiring substantial cooling and heat input. This makes CO_2 capture one of the major operating costs in typical plants. To reduce this energy burden, our process incorporates a mixture of substituted amines that enables the absorbent solution to spontaneously form two liquid phases during CO_2 absorption: a CO_2-rich phase and a CO_2-lean phase. The CO_2-rich phase is denser, and the two phases become immiscible, allowing for separation by gravity. Only the CO_2-rich phase is sent to the regenerator, significantly lowering the required heat duty. In contrast, traditional single-phase absorbent systems send the entire absorbent stream to regeneration, leading to much higher energy consumption.

Other areas of our overall process also demonstrate energy efficiency. In the MTO section, the fluidized-bed reactor that converts MeOH to light olefins maintains its temperature by receiving heat from the catalyst regenerator. The regenerator is directly connected to the reactor to enable catalyst recirculation, and during regeneration, when air reacts exothermically with the SAPO-34 catalyst, the released heat is transferred directly to the reactor. This reduces the external energy required to maintain the optimal reactor temperature for achieving the desired selectivity and conversion.

A final example of energy efficiency in our project is found in the CO_2-to-MeOH synthesis section. Using Aspen's Energy Analyzer v11, a heat-exchanger network (HEN) is developed that significantly improved energy integration and reduced utility consumption. By strategically incorporating three heat exchangers into the final design, overall energy savings increased by 63.2%. External hot-utility demand decreased from 3.13×10^{1} to 1.4×10^{-2} GW (a 99.6% reduction), and external cooling-utility demand decreased from 3.62×10^{1} to 2.11×10^{1} GW (a 41.7% reduction). (Yusuf and Almomani 2023)

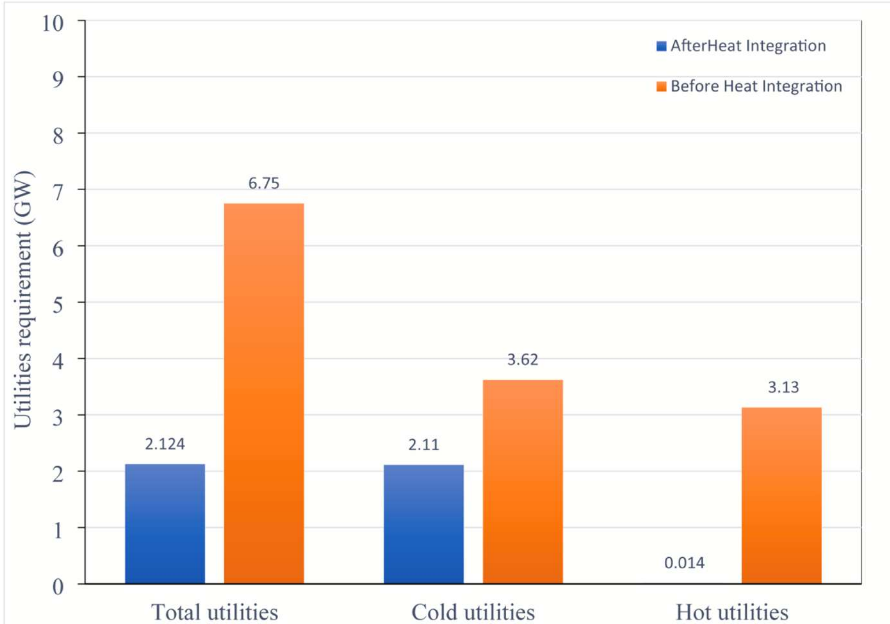

Figure 5. Required External Utility Before and After the Optimized H.E.N (Yusuf and Almomani 2023)

One additional factor that enhances the appeal of our project is the availability of federal carbon management incentives. Although Indiana does not impose a state-level carbon tax on industrial facilities, the federal 45Q tax credit and provisions within the Inflation Reduction Act (IRA) apply to qualifying operations. These programs offer substantial financial incentives for facilities engaged in carbon capture or utilization, including those located in Indiana. In summary, the potential for federal tax credits, the environmental advantages of our process compared to similar technologies, and the efficient use of energy collectively make our project a strong and viable endeavor. (Jones 2023)

4 Process Description

4.1 Chosen Process

The overall process is designed as an integrated pathway that converts natural gas and captured carbon dioxide into high-value light olefins, primarily ethylene and propylene, through a series of interconnected reaction and separation steps. The process begins with hydrogen production via steam methane reforming (SMR), where natural gas is converted into a hydrogen-rich synthesis gas, followed by water-gas shift and purification to produce high-purity hydrogen. In parallel, carbon dioxide is captured from process streams, providing a concentrated CO_2 source for utilization rather than emission. These feedstocks are combined in the methanol synthesis section, where CO_2 and H_2 are catalytically hydrogenated over a $Cu/ZnO/Al_2O_3$ catalyst to produce methanol at high pressure and moderate temperature. The produced methanol is then converted into light olefins in a fluidized-bed methanol-to-olefins (MTO) reactor–regenerator system, where continuous catalyst circulation enables high conversion and sustained activity. Following reaction, high-pressure CO_2 capture and molecular sieve purification are implemented to remove impurities and recycle CO_2 upstream, reinforcing a closed carbon loop. Finally, the hydrocarbon mixture is separated through membrane and cryogenic distillation systems to recover polymer-grade ethylene and propylene. This fully integrated process emphasizes efficient energy use, advanced separation technologies, and carbon management strategies to achieve sustainable, economically viable olefin production.

4.1.1 H_2 Production and CO_2 Capture from Flue Gas Emissions

4.1.1.1 Hydrogen (H_2) Production Process

Hydrogen production for this plant will use steam methane reforming (SMR). SMR is a widely used industrial process that converts natural gas into high-purity hydrogen through a series of catalytic reactions and separations. The process is designed to maximize hydrogen yield while maintaining high thermal efficiency through heat integration within the plant.

The process begins with natural gas purification to remove sulfur compounds and other contaminants that could interfere with catalysts used later in the process. The purified natural gas is typically supplied at pressures of 20 to 30 bar, ensuring compatibility with subsequent high-pressure processing units.

Following purification, natural gas is mixed with steam and introduced into the primary reformer. This reactor operates at high temperatures between 800 and 900°C and at moderate pressures of approximately 20 to 30 bar. Within the reformer, methane reacts with steam over a nickel-based catalyst according to the endothermic reaction for the process of producing hydrogen:

$$CH_4 + H_2O \rightarrow CO + 3H_2$$

The reformer consists of an external heat source, where heat is supplied by combustion in the furnace. The high temperature favors hydrogen production, while the operating pressure is selected to balance reaction kinetics and downstream separation requirements. The resulting synthesis gas (syngas) contains hydrogen, carbon monoxide, carbon dioxide, and small amounts of unreacted methane.

The syngas is then directed to the high-temperature shift (HTS) reactor, which operates at approximately 20 to 25 bar. In this part of the SMR process, carbon monoxide reacts with steam to produce additional hydrogen and carbon dioxide via the water gas shift reaction: (IEA Greenhouse Gas R&D Programme (IEAGHG) 2017)

$$CO + H_2O \rightarrow CO_2 + H_2$$

This step significantly increases the hydrogen yield by converting most of the carbon monoxide into hydrogen. The outlet stream from the shift reactor is therefore rich in hydrogen and carbon dioxide, with only small amounts of residual carbon monoxide remaining.

Subsequently, the gas mixture enters a pressure swing adsorption (PSA) unit for hydrogen purification. The PSA system operates at adsorption pressures of 20 to 30 bar, and the regeneration pressure is near 1 bar. Within the PSA, hydrogen is selectively separated from impurities such as CO_2, CO, CH_4, and nitrogen using cyclic adsorption beds. This process produces hydrogen with a purity of 99.9%, while achieving a recovery efficiency of approximately 85 to 90%. The off gas from the PSA, known as tail gas, is recycled and used as fuel in the reformer furnace, thereby improving overall process efficiency.

An important feature of the SMR process is its extensive heat integration. Heat recovered from the reformer flue gas and hot process streams is used to generate high-pressure steam, which is recycled within the process and supports the reforming and shift reactions.

The final hydrogen product is delivered at approximately 25 bar and 40°C, meeting stringent industrial purity specifications. Carbon monoxide and carbon dioxide concentrations are typically limited to less than 10 ppm, ensuring suitability for developing methanol-to-olefin production.

Overall, the SMR-based hydrogen production process combines catalytic reforming, equilibrium-shift reactions, and advanced gas separation techniques to efficiently produce high-purity hydrogen for industrial scale use.

4.1.1.2 Carbon Dioxide Capture Process from SMR Flue Gas

Carbon dioxide (CO_2) capture in an SMR-based hydrogen production plant is implemented to reduce greenhouse gas emissions by separating CO_2 from process gas streams prior to release. The process can be integrated within the plant, depending on the selected capture strategy, with the most common approaches involving capture from shifted syngas.

The most widely applied and commercially mature method is pre-combustion capture from shifted syngas. After the water-gas shift reactor, the gas stream contains a high concentration of CO_2 at an elevated pressure of 20 to 25 bar. This high partial pressure makes the stream particularly suitable for chemical absorption. In this process, the gas is contacted with a liquid solvent, typically an aqueous solution of methyl diethanolamine (MDEA), in an absorption column. The CO_2 is selectively absorbed into the solvent, while hydrogen-rich gas exits the top of the column for further purification. (de Riva, et al. 2018)

The CO_2-rich solvent is then sent to a regeneration (stripper) column, where it is heated to release the absorbed CO_2. This regeneration step typically operates near atmospheric pressure, which is approximately 1 to 2 bar, and requires significant thermal energy, usually supplied by steam extracted from the plant's heat recovery system. The regenerated solvent is recycled back to the absorber, forming a continuous loop.

Regardless of the capture location, the separated CO_2 stream undergoes compression and dehydration to meet transportation and storage specifications. The CO_2 is compressed to pipeline conditions, typically around 110 bar, and dried to remove moisture, preventing corrosion and hydrate formation during transport. The final CO_2 product has a purity of approximately 99% and is suitable for pipeline transport, geological storage, or utilization in enhanced oil recovery.

The integration of CO_2 capture systems into the SMR process increases energy demand, particularly for solvent regeneration and gas compression. This results in increased fuel consumption and reduced net power exports from the plant. However, the implementation of carbon capture technologies enables significant reductions in CO_2 emissions, with overall capture rates ranging from approximately 50% to 90%. (IEA Greenhouse Gas R&D Programme (IEAGHG) 2017)

4.1.2 Methanol Synthesis

The chosen process converts carbon dioxide (CO_2) and hydrogen (H_2) into methanol (CH_3OH) through catalytic hydrogenation. Four major steps take place: Feed Preparation, Methanol Synthesis, Separation, and Purification.

The process begins by preparing and conditioning the feed streams before they enter the catalytic reactor. Carbon dioxide and hydrogen are supplied as different feeds and are initially compressed to the required reaction pressure. They are then mixed, and this combined stream forms the stoichiometric CO_2/H_2 ratio necessary for methanol synthesis. The stream is then routed through a heat exchanger that recovers heat from hot process streams. Exchanges raise the feed temperature, reducing the energy required later in the heater. Finally, an external heater raises the stream temperature to approximately 210°C, ensuring the mixture reaches the reactor at optimal reaction conditions.

One conditioned, the feed enters the methanol synthesis reactor, a catalytic reactor packed with $Cu/ZnO/Al_2O_3$ catalyst. This catalyst system is widely used in industrial methanol production and is well supported by various studies. Within the reactor, CO_2 and H_2 undergo exothermic hydrogenation reactions to form methanol and water. The primary reactions are the following:

$$CO_2 + 3H_2 \rightarrow CH_3OH + H_2O$$

$$CO + 2H_2 \rightarrow CH_3OH$$

In addition, the reverse water-gas shift (RWGS) reaction also takes place:

$$CO_2 + H_2 \rightarrow CO + H_2O$$

Although RWGS introduces an additional reaction pathway, CO is also consumed in methanol formation, and kinetic studies have shown that CO_2 remains the main carbon source for methanol synthesis under these conditions. The reactor operates at high pressure and moderate temperature to maximize conversion, in accordance with the design conditions described by Yusuf et al.

The hot reactor effluent exits and immediately enters a cooling and condensation train. Heat exchangers remove heat from the reaction products, lowering the temperature enough to condense crude methanol and water from the gas mixture. The cooled two-phase stream flows into the first separation drum, where liquids and gases are separated. The liquid contains crude methanol, water, dissolved CO_2, and trace components, while the vapor stream consists primarily of unreacted CO_2 and H_2, with small amounts of CO. To improve overall conversion, the vapor stream is recycled back to the feed system. A small purge is taken to prevent the accumulation of inert.

The crude liquid undergoes additional cooling and flashing through an exchanger and a separation drum to remove remaining light gases. After gas removal, the liquid is preheated and directed to the purification section. A distillation column separates the methanol-water mixture into a high-purity methanol and a water-rich bottoms stream. The overhead vapor is condensed and collected as the final methanol product. Overall, the process integrates compression, heat recovery, catalytic reaction, and separation into a continuous process that converts CO_2 to methanol, achieving a methanol purity of approximately 99%. (Yusuf and Almomani 2023)

The operating conditions for the CO_2-to-methanol process were selected based on published kinetic and process studies to achieve high conversion and efficient system performance. The CO_2 feed enters at 12.3°C and 47.63 bar, while hydrogen enters at 25°C and 30 bar. Both streams undergo compression so that the final mixed feed reaches a pressure of approximately 75-78 bar before entering the reactor. Operating at this elevated pressure is essential because the methanol synthesis reactions involve a decrease in volume, and high pressure shifts the equilibrium toward methanol formation. Before entering the reactor, the feed mixture is heated to around 210°C using a combination of heat recovery exchangers and heaters. This temperature is consistent with the optimal range for Cu/ZnO/Al2O3 catalysts, which show strong activity between 200 and 260°C. (Yusuf and Almomani 2023)

Inside the reactor, the system operates at an inlet pressure of 75.7 bar and a controlled temperature of 210°C. These values match the specifications from the referenced study, which demonstrated that these conditions yield approximately 96% CO_2 conversion and 99% methanol production under isothermal operation (Reference). The reactor contains a large mass of Cu/ZnO/Al2O3 catalyst (44,500 kg), and a pressure drop of about 0.6 bar occurs across the reactor.

Following the reaction, the effluent enters the separation and purification stages. The first drum operates near reactor pressure (approximately 73-74 bar) to condense crude methanol without excessive vaporization. Subsequent flash drums operate at reduced pressures of 1-2 bar to remove dissolved gases and prepare the liquid mixture for distillation. The methanol purification column operates close to atmospheric pressure with a reflux ratio of 2.12. Temperatures within the column typically range between 64°C and 102°C, depending on stage composition. The condenser cools the overhead product to around 14.95°C, allowing for the collection of high purity methanol. These operating conditions allow the process to achieve stable operation, high conversion, and efficient energy use.

4.1.3 Methanol to Olefins

The methanol-to-olefins (MTO) process converts methanol (CH_3OH) into olefin hydrocarbons, primarily ethylene (C_2H_4) and propylene (C_3H_6), through catalytic reactions over a SAPO-34 molecular sieve catalyst. The process is implemented using a fast fluidized-bed reactor coupled with a regenerator, allowing continuous catalyst circulation and sustained activity despite coke formation.

Methanol is first pressurized and preheated using heat integration with hot process utility streams before entering the reactor to meet reactor operating conditions. Heat integration is also implemented to reduce external heating requirements. The preheated methanol is vaporized and introduced at the base of the fluidized bed reactor. The SAPO-34 catalyst is maintained in a fast fluidized state, which provides gas-solid contact, high heat transfer rates, and uniform temperature distribution. (Yuan 1996)Under these conditions, methanol undergoes a series of catalytic transformations governed by the hydrocarbon pool mechanism, where intermediate hydrocarbon species within the catalyst pores facilitate chain growth and cracking reactions. Within the reactor, methanol undergoes a complex reaction network that produces light olefins, water, and secondary heavy hydrocarbons. The dominant reactions can be presented as:

$$CH_3OH \rightarrow C_2H_4 + H_2O$$

$$CH_3OH \rightarrow C_3H_6 + H_2O$$

In addition to these primary pathways, side reactions generate methane, ethane, butanes, pentenes, carbon monoxide, and hydrogen. Additionally, a portion of the methanol is converted into coke, which is deposited on the catalyst surface.

To maintain continuous operation, the process incorporates a catalyst regeneration loop. A key challenge in the MTO process is catalyst deactivation due to coke formation, which deposits on the catalyst surface and reduces activity. To address this, the process incorporates a regenerator where coke is combusted with oxygen to form CO_2 and H_2O, restoring catalyst performance and enabling continuous operation. Spent catalyst containing coke is continuously withdrawn from the reactor and transferred to the regenerator. In the regenerator, air is introduced, and the coke is combusted through oxidation reactions:

$$C + O \rightarrow CO_2$$

$$H + \frac{1}{4}O_2 \rightarrow \frac{1}{2}H_2O$$

These reactions remove the deposited carbon and hydrogen species, restoring the catalyst's active sites. The regenerated catalyst, now at elevated temperature due to the exothermic combustion reactions, is circulated back to the reactor, where it supplies part of the heat required for the endothermic methanol conversion reactions. This thermal coupling between reactor and regenerator significantly improves energy efficiency and reduces the need for external heating.

The reactor system operates at approximately 775 K and 4.5 bar, conditions identified as optimal for maximizing olefin yield while maintaining catalyst stability. Under these conditions, methanol conversion exceeds 99.9% while combined ethylene and propylene yield approximately 30%.

The fluidized-bed configuration plays a critical role in process performance. It ensures rapid mixing of gas and solid phases, minimizes temperature gradients, and prevents localized hot spots that could accelerate catalyst deactivation. Additionally, continuous catalyst circulation allows the system to operate at steady state with consistent activity, overcoming one of the primary limitations of fixed-bed reactor systems in MTO applications.

Overall, this reactor–regenerator system integrates reaction kinetics, catalyst management, and heat transfer into a single continuous process, enabling high conversion efficiency, controlled selectivity toward light olefins, and sustained industrial operation. (UOP 2019)

4.1.4 High Pressure CO_2 Capture

The effluent stream exiting the methanol-to-olefins (MTO) reactor contains a multicomponent mixture containing light olefins, heavy olefins, and byproducts. The byproducts include carbon dioxide (CO_2), which needs to be removed to 500 ppm before moving upstream to the cryogenic separation units. This removal also gives the opportunity to further implement the environmentally friendly motive, as the captured carbon can be used upstream as a CO_2 inlet.

Following initial cooling and bulk CO_2 removal through high-pressure absorption, a final polishing and drying step is required to ensure that the gas stream meets the stringent specifications necessary for cryogenic separation. This is achieved through the use of a molecular sieve adsorption system, which plays a critical role in removing residual CO_2 and trace water to very low concentrations.

The molecular sieve used in this process is typically a 3A zeolite, selected based on its pore size (~3 Å), which allows for selective adsorption of small polar molecules such as water while excluding larger hydrocarbon molecules like ethylene and propylene. The adsorption mechanism is based on physical adsorption driven by electrostatic interactions, where polar molecules are attracted to the charged internal surfaces of the crystalline aluminosilicate structure. Because water has a strong dipole moment, it is preferentially adsorbed even at very low partial pressures, making molecular sieves highly effective for deep dehydration. (ZR Catalyst n.d.)

In addition to water removal, molecular sieves can also contribute to the removal of residual CO_2, particularly when concentrations have already been reduced by upstream absorption units. CO_2 molecules, due to their quadrupole moment, also interact strongly with the adsorbent surface and can be captured within the pore structure. This dual functionality ensures that both water and CO_2 are reduced to levels that prevent operational issues in downstream units.

The adsorption system is typically designed as a multi-bed configuration operating in cyclic mode, consisting of at least two parallel beds. While one bed is in adsorption mode, removing impurities from the process stream, the other undergoes regeneration. This allows for continuous operation without interrupting the process flow. During adsorption, the gas stream passes through the packed bed of molecular sieve material, where impurities are retained and a purified hydrocarbon stream exits.

Regeneration of the saturated bed is achieved through a temperature swing adsorption (TSA) process, where a heated inert or process gas (often a slipstream of dry hydrocarbon gas) is passed through the bed to desorb the retained species. The elevated temperature weakens the adsorption forces, releasing water and CO_2 from the pores. The desorbed impurities are then removed from the system, and the bed is cooled before being returned to service. This cyclic operation ensures long-term usability of the adsorbent and maintains consistent product quality.

The implementation of molecular sieves is particularly critical for cryogenic separation systems, where temperatures can drop below −100°C. At these conditions, even trace amounts of water or CO_2 can freeze and form solid deposits, leading to blockages, pressure drops, and potential equipment failure in distillation columns and heat exchangers. By reducing water content to parts-per-million levels and polishing residual CO_2, the molecular sieve unit ensures reliable and safe operation of the downstream separation train. (de Riva, et al. 2018)

Overall, the molecular sieve system serves as the final purification barrier, complementing upstream high-pressure CO_2 absorption by achieving ultra-low impurity levels. This ensures compatibility with cryogenic processing while also enabling efficient recovery and reuse of CO_2 within the integrated process, reinforcing both operational efficiency and environmental sustainability.

4.1.5 Olefin Separation

Following the initial CO_2 and water removal, the resulting hydrocarbon stream consists primarily of ethylene, propylene, methane, ethane, butene, pentenes, and residual hydrogen. Due to the wide range of molecular weights, boiling points, and non-ideal phase behavior, a multi-stage separation strategy combining membrane separation and cryogenic distillation is required to achieve polymer-grade olefins.

The first step of the purification if a hydrogen removal using palladium-based membrane separation. Hydrogen is present as a by-product of several MTO reactions, including methanol decomposition and water-gas shift pathways. If not removed, hydrogen would accumulate in downstream distillation columns, reducing relative volatilities and significantly increasing energy consumption. The palladium membrane selectively permeates hydrogen due to its ability to dissociate molecular hydrogen into atomic hydrogen, which diffuses through the metal lattice.

The hydrogen-depleted hydrocarbon stream then enters the cryogenic distillation train, which is designed as a sequence of columns arranged by carbon number. The separation is carried out under progressively decreasing pressures to balance vapor-liquid equilibrium behavior and minimize refrigeration and compression requirements.

The first column in the sequence operates at high pressure and performs the C_1/C_2 split. In this column, methane and any remaining light gases are separated from the heavier C_2^+ components. Operating at elevated pressure increases condensation temperatures, reducing refrigeration duty and improving energy efficiency. The overhead stream, rich in methane, is typically routed for fuel use within the plant. The bottoms product, containing the heavier hydrocarbons, proceeds to the next stage.

The second column focuses on C_2 separation, where ethylene is separated from ethane. This separation is challenging due to the similar boiling points and relative volatility of ethylene and ethane. As a result, the column operates under cryogenic conditions with a high number of theoretical stages and a significant reflux ratio to achieve the polymer-grade ethylene. Ethane is removed as the bottom's product.

The third column operates at an intermediate pressure of approximately 18 bar and performs the C_3 split, separating propylene from propane. Similar to the C_2 splitter, this column requires careful control of reflux ratio and temperature profiles due to the close volatilities of the components. The overhead stream yields high-purity propylene suitable for polymerization, while propane is removed as a bottoms stream. (UOP 2019)

The remaining heavier hydrocarbons (C_4 and above) are processed in lower-pressure columns (~5 bar) to separate butenes, pentenes, and heavier fractions. Lower operating pressures are used in this section to facilitate separation of heavier components without excessive reboiler duty. These streams are typically less valuable than ethylene and propylene but can still be utilized as fuel, blended into chemical feedstocks, or further separated if economically justified.

A key design feature of this separation train is the cascading pressure configuration, where higher pressures are used for lighter components and lower pressures for heavier fractions. This approach optimizes energy usage by reducing refrigeration requirements in the front-end separations and minimizing reboiler duty in the heavier separations. Additionally, it allows for better thermal integration across the system, where heat removed from one unit can be reused in another. (UOP 2019)

To further improve energy efficiency, the process incorporates heat integration through a network of heat exchangers designed using pinch analysis. Waste heat from reactor effluent and exothermic processes is reused to supply reboiler duties or preheat feed streams, while cold streams from cryogenic units are used to condense overhead vapors. This significantly reduces external utility requirements and improves overall process sustainability. (UOP 2019)

Overall, the olefin separation section integrates membrane technology, cryogenic distillation, pressure optimization, and heat recovery to efficiently produce polymer-grade ethylene and propylene. The system is designed to handle a complex multicomponent mixture while minimizing energy consumption and maximizing product recovery, making it a critical component of the overall MTO process.

4.1.6 Competing Processes Analysis

4.1.6.1 CO_2 to Gasoline

An alternative process considered for the plant development is the conversion of carbon dioxide (CO_2) to gasoline-range hydrocarbons. The process involves a multi-step catalytic process that enables the production of liquid transportation fuels from captured carbon. In the process, CO_2 is reacted with hydrogen (H_2) by electrolysis to form carbon monoxide (CO) via the reverse water-gas shift reaction. The resulting synthesis gas CO + H_2 is then fed into a Fischer–Tropsch synthesis reactor, where it is converted over a cobalt-based catalyst at temperatures between 200 and350 °C and pressures of 10 to 40 bar into a mixture of hydrocarbons and water. (James Robert Jennings 2012)

The primary reactions of CO_2 to Gasoline:

$$CO_2 + H_2 \rightleftharpoons CO + H_2O$$
$$CO + H_2 \rightarrow \text{Gasoline} + H_2O$$

Overall, the CO_2-to-gasoline process represents an alternative route for carbon recycling and sustainable fuel production. The market value of gasoline is \$ 1,535 per ton. (Petroleum & Other Liquids Prices 2026) The alternative process of converting CO_2 to Gasoline involves additional regulations on gasoline storage and handling, which increase the cost and complexity of operating the plant. The refining of gasoline involves a larger process that requires more equipment and utilities, leading to higher development costs.

4.1.6.2 CO_2 to Methanol

An alternative design approach is to rely solely on the catalytic hydrogenation of carbon dioxide (CO_2) to methanol (CH_3OH) for carbon utilization and sustainable fuel production. In the process, captured CO_2 is compressed and purified before being reacted with hydrogen (H_2), typically produced via water electrolysis or steam reforming. The reaction is carried out over a copper-based catalyst $Cu/ZnO/Al_2O_3$ in a fixed-bed reactor at temperatures between 200 and 300 °C and pressures at 50 to 100 bar. (Jennings 2014)

The primary reaction in the process is:

$$CO_2 + 3H_2 \rightleftharpoons CH_3OH + H_2O$$

Methanol can be sold as an industrial-grade chemical for further synthesis into more complex chemicals, such as polymers and gasoline. The market value of methanol is $1247 per ton. (Methanex Corporation 2026) The concept of operating the plant to produce methanol lowers development and utility costs but comes at the expense of higher utility costs during operating hours and increased expenditures for handling and storing methanol.

4.1.6.3 Comparison of Competing Processes

The selected process for the plant design basis is CO_2 Carbon Capture to Olefin Monomer production. The selected process was chosen for the following factors, which made the plant's development more feasible than the other two methods. While CO_2 to Methanol is simpler to produce than monomers, the value of selling methanol does not justify the plant's development in terms of profitability. In the case of CO_2 to gasoline, the complexity of refining CO_2 into gasoline increases operational costs, exceeding the profits from selling gasoline.

4.1.6.4 Table 3: Comparison of Competing Processes

Process	Market value of product in USD:	Pros	Cons
CO_2 To Methanol	$1247 per ton	Simple process to develop.	Low market value of methanol.
CO_2 To Olefins	Propylene: $1270 per ton Ethylene: $520 per ton	Multiple products are produced.	Limited to the monomer market.
CO_2 Gasoline	$3.31 per gallon	Higher market value in selling the product.	Complexity in producing gasoline leads to high operating costs.

5 Process Control

5.1 Process Control Objectives

The main control objective of the process is to maintain a safe and consistent operation of the method of converting carbon dioxide captured from pipelines and hydrogen to standard-grade ethylene and propylene. The process involves two main phases: CO_2 hydrogenation to methanol and methanol-to-olefins (MTO) with olefin separation and purification. The full control plan for the plant must focus on maintaining stable and consistent feed delivery, ideal reaction conditions and dependable separation performance while integrating the utilities and recycle streams throughout the whole process.

Since the CO_2 capture system and the H_2 production phase considered as black boxed in our process, the main control goal in the plant is to maintain stable CO_2 and H_2 pressures, compositions and feed flow rates before entering the methanol synthesis section. Though the upstream units aren't controlled in the process outline, their outlets conditions are a strong influence on the downstream separations and reactors. The first phase of control is to ensure the feed streams entering the methanol synthesis section remain as stable as possible so possible disturbances from the black boxed units don't spread throughout the rest of the process.

The first major control goal is maintaining appropriate operating conditions in the methanol synthesis section. For this phase of the process, the compressed carbon dioxide and hydrogen streams are mixed then preheated before entering the methanol reactor. A stable reactor needs maintain the appropriated H_2/CO_2 feed ration, reactor inlet temperature and a controlled pressure. These variables affect reaction conversion, stability and catalyst performance. The heating and heat-exchange network must regulate the feed temperature to ensure the reactor gets the appropriate feed stream.

For the downstream of the methanol reactor, the objective moves to maintaining stable separation and recycle behavior. The reactor effluent is cooled then sent through the separation and flash vessels that remove the condensed methanol and water from the unreacted gases. Temperature, pressure, and liquid level control are required to ensure a reliable phase separation and stop disturbances from occurring in the recycle loop. Appropriate recycle management is important for improving the conversion while stopping excessive buildup of the unreacted gases or other inert components in the system. The methanol purification must maintain steady operator for production of a consistent methanol feed stream in the MTO process.

The second control goal happens in the methanol-to-olefins (MTO) section where the methanol is converted to hydrocarbons. The MTO reactor must maintain stable pressure, temperature, and feed flow conditions to achieve a steady methanol conversion and good selectivity toward ethylene and propylene. Since the MTO reaction is very temperature-sensitive and involves regeneration of a catalyst, the reactor temperature control is important to stop formation of heavier hydrocarbons or catalyst deactivation. The reactor ensures that the separator units are supplied with a steady hydrocarbon mixture.

After the MTO reactor, the product stream goes through compression, cooling, absorption, and distillation to separate methane, ethylene, propylene, and other heavier hydrocarbons. The main control goal in this section is achieving stable separation performance and meeting the polymer-grade product specs needed. This is achieved through maintaining the column pressures, reboiler duties, reflux rations, and liquid inventory levels in the distillation column. Since these columns operate in a series, the disturbances in one separation unit can spread to the downstream units, making a coordinated temperature and pressure control important in the system.

Throughout the plant, a unified control strategy must maintain control stability and energy balance. Steam systems, heat exchangers, cooling water, and refrigeration units need to function effectively to support the MTO section. Effective utility managements Is essential for temperature control in reactors and separation units' ability to remain stable during load changes.

Ultimately, the control objectives implemented in the plant are to maintain consistent feed delivery and appropriate methanol synthesis conditions, attain a consistent olefin conversion in the MTO reactor, and maintain a steady operation of the downstream hydrocarbon separation system. Achieving the objectives for a fully integrated process to efficiently and safely operate while achieving maximum olefin production from the capture dioxide.

5.2 Process Control Loop Over Methanol Synthesis Reactor

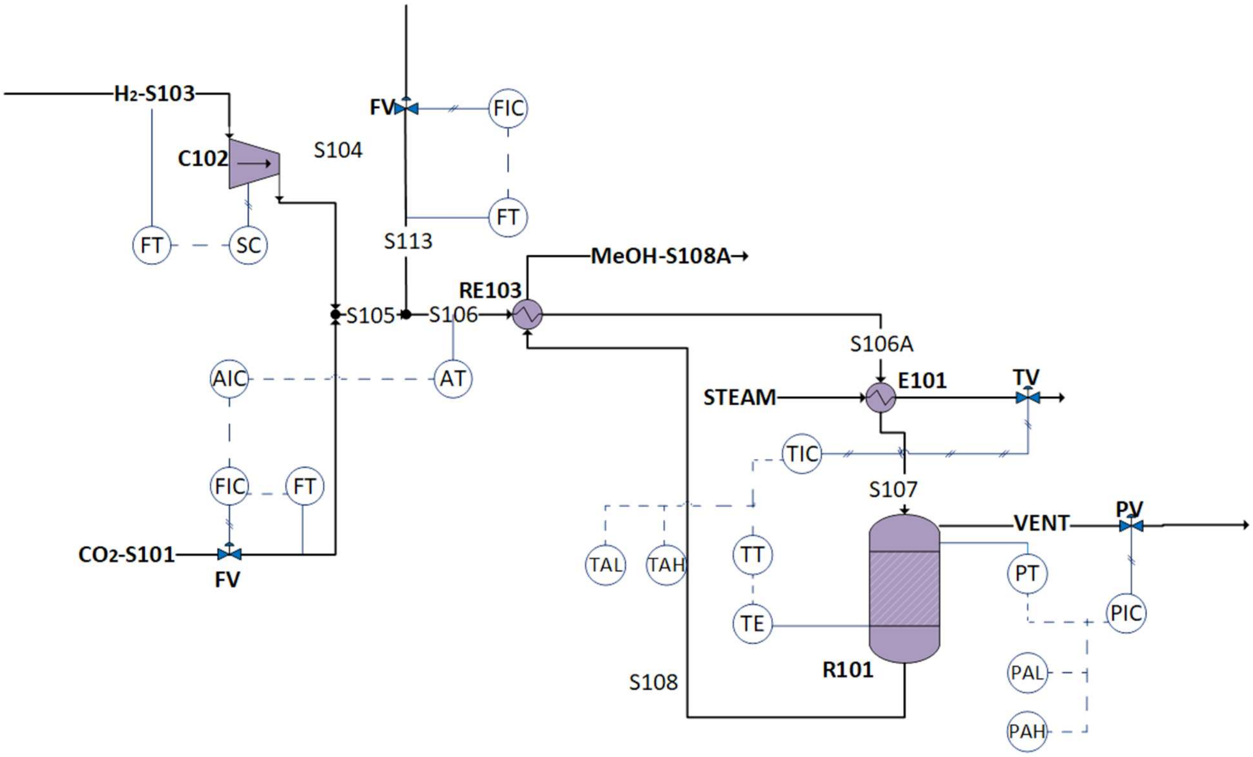

Figure 6: Process Control Loop for Methanol Synthesis Reactor

The control system shown in Figure 6 regulates the flow of hydrogen and carbon dioxide into the reactor and maintains stable reactor operating conditions. At the inlet, the flow of hydrogen (H_2-S103) is controlled using a speed controller (SC) that adjusts the compressor output to keep the hydrogen supply consistent.

The carbon dioxide flow (CO_2-S101) is also controlled using a flow controller (FIC) and control valve, which ensures that the correct amount of CO_2 enters the system. An analyzer (AT) measures the composition of the mixed stream (S106) before entering the reactor and it helps maintain a 1:7 ratio of CO_2 to H_2 by adjusting the carbon dioxide inlet flow rate as needed.

In the reactor, temperature is controlled using a temperature controller (TIC) that regulates the flow of steam through the heat exchanger E101. This helps remove excess heat and keeps the reactor temperature stable. Temperature alarms (TAL and TAH) are also included to alert operators if temperature becomes too high or too low.

Pressure in the reactor is controlled by using a pressure controller (PIC) that adjusts a valve on the vent line. This prevents pressure from getting too high. Pressure alarms (PAH and PAL) are included to warn if pressure goes outside safe limits.

6 Process Safety and Environmental Analysis

6.1 Air Emissions

Air emissions from the process primarily consist of carbon dioxide (CO_2), carbon monoxide (CO), hydrogen (H_2), methane (CH_4), and light hydrocarbons such as ethylene (C_2H_4), propylene (C_3H_6), butene (C_4H_8), ethane (C_2H_6), and pentenes (C_5H_{10}). These gases are generated in several process streams, including S111, S117, S140, S151B, S151C, S151D, and S159. Because many of these compounds are volatile hydrocarbons that contribute to air pollution and ground-level ozone formation, emissions must be carefully managed to prevent uncontrolled atmospheric release.

Air waste from the process is typically controlled through several mitigation systems. Hydrocarbon gases and methanol vapors can be routed to flare systems or thermal oxidizers where they are combusted under control conditions, converting them primarily to carbon dioxide and water. This reduces the release of unburned hydrocarbons into the atmosphere. Additionally, closed processing systems and vapor recovery units may be used to capture and recycle valuable gases such as hydrogen or propylene back into the process, reducing both emissions and material losses. Methane and other hydrocarbon gases may also be used as supplemental fuel in process heaters or boilers when recovery is feasible. (U.S. Environmental Protection Agency (EPA) n.d.)

Air emissions from industrial facilities are regulated primarily under the Clean Air Act (CAA). This legislation requires facilities to obtain operating permits and implement emission control technologies to limit the release of regulated air pollutants. Hydrocarbon emissions are controlled because they contribute to photochemical smog formation, while carbon monoxide emissions must be minimized due to their effects on air quality and human health. Facilities that emit significant quantities of greenhouse gases, such as carbon dioxide or methane, may also be subject to reporting requirements under the EPA Greenhouse Gas Reporting Program (40 CFR Part 98). Continuous monitoring and proper emission-control systems ensure that air waste from the process remains within regulatory limits and minimizes environmental impacts. (Resource Conservation and Recovery Act Hazardous Waste Regulations n.d.)

Table 4: Vapor Waste Summary Table

Substance	**Total Mass Flow (ton/year)**	**Streams**	**Waste Type**	**Waste Management**	**EPA Regulation**
CO (Carbon Monoxide)	51,174.00	S-111, S-117, S-140	Process gas	Control through combustion.	Regulated under the Clean Air Act National Ambient Air Quality

					Standards for carbon monoxide emissions.
Oxygen (O_2)	179,193	S-130	Process gas	Reused in the process.	Regulated under the Clean Air Act.
Nitrogen (N_2)	924,733	S-130	Process gas	Safely vented or recycled where possible; maintain ventilation to prevent oxygen displacement.	Regulated under the Clean Air Act.
CO_2 (Carbon Dioxide)	78,116.00	S-111, S-117, S-140	Process gas	Vent through controlled emission systems.	Regulated under the Clean Air Act and the EPA Greenhouse Gas Reporting Program (40 CFR Part 98) .
H_2 (Hydrogen)	291,999.00	S-111, S-117	Process gas	Recycle within the process when possible.	Follow industrial safety regulations.

6.2 Liquid Emissions

Several process streams contain light hydrocarbons including C_3 hydrocarbons, propylene (C_3H_6), ethylene (C_2H_4), butene (C_4H_8), methane (CH_4), ethane (C_2H_6), and pentane (C_5H_{10}). These materials primarily appear in streams S159, S151B, S151C, and S151D. Because these compounds are volatile hydrocarbons that can contribute to atmospheric pollution and ground-level ozone formation, they must be handled using closed processing systems to prevent uncontrolled emissions. In most cases, these hydrocarbons are either recycled back into the process, recovered for use as fuel, or directed to controlled combustion systems such as flares. This approach reduces atmospheric emissions while allowing for recovery of the energy value of hydrocarbons. Emissions of hydrocarbon gases are regulated under the Clean Air Act, which requires facilities to maintain air permits and operate emission-control equipment to limit atmospheric releases (Resource Conservation and Recovery Act Hazardous Waste Regulations n.d.)

Large quantities of oxygen (O_2) and nitrogen (N_2) are also present in the process through stream S130. These gases are typically stored and transported as cryogenic liquids but rapidly vaporize under ambient conditions. Although oxygen and nitrogen are not considered hazardous environmental pollutants, improper release can create operational safety hazards such as oxygen enrichment or oxygen displacement in confined spaces. These gases are therefore typically vented through controlled ventilation systems or reused within the process where possible. Because they are not classified as hazardous waste, they are not regulated under the Resource Conservation and Recovery Act (RCRA); however, proper storage and handling procedures must follow industrial safety standards and workplace safety guidelines (U.S. Environmental Protection Agency (EPA) n.d.)

Water represents the largest material flow in the process, with approximately 3.79 million tons per year passing through streams S151B, S151C, S151D, and S153. Additional smaller quantities of water are present in synthesis streams such as S111 and S117. Process water may contain dissolved organics such as methanol or trace hydrocarbons, which require treatment before discharge. Wastewater generated in the process is collected and routed to a wastewater treatment system where contaminants are removed through processes such as filtration, neutralization, and biological treatment.

Before discharge, the treated wastewater must meet limits established by the National Pollutant Discharge Elimination System (NPDES) under the Clean Water Act. These regulations establish allowable pollutant concentrations for wastewater discharged to municipal treatment facilities or surface waters. Monitoring and treatment ensure that wastewater released from the facility does not interfere with public wastewater treatment systems or negatively affect aquatic ecosystems (Resource Conservation and Recovery Act Hazardous Waste Regulations n.d.)

Table 5: Liquid Waste Summary Table					
Substance	**Total Mass Flow (ton/year)**	**Streams**	**Waste Type**	**Simplified Waste Management**	**EPA Regulation**

Ethylene (C_2H_4)	1.4200	S-151B, S-151C, S-151D	Liquid hydrocarbon waste	Recover for further use.	Regulated as volatile organic compound emissions under the Clean Air Act.
C_4H_8 (Butene)	0.5550	S-159, S-151B, S-151C, S-151D	Liquid hydrocarbon waste	Store in sealed containers and recycle.	Regulated as volatile organic compound emissions under the Clean Air Act.
CH_4 (Methane)	0.0658	S-151B, S-151C, S-151D	Hydrocarbon in dissolved in liquid	Recycle as fuel.	Regulated under the EPA Greenhouse Gas Reporting Program and Clean Air Act methane emission rules.
H_2 (Liquid Hydrogen)	0.0217	S-151B, S-151C, S-151D	Process Gas dissolved in liquid	Reuse in process.	Regulated under the Clean Air Act.
Ethane (C_2H_6)	0.0086	S-151B, S-151C, S-151D	Liquid hydrocarbon waste	Recover as fuel.	Regulated under the Clean Air Act hydrocarbon emission regulations.
C_5H_{10} (Pentene)	0.0001	S-159, S-151B, S-151C, S-151D	Liquid hydrocarbon waste	Store in sealed containers and recycle.	Regulated as volatile organic compound emissions under the Clean Air Act.

Propane (C_3H_8)	3,262.87	S-159, S-151B, S-151C, S-151D	Liquid hydrocarbon waste	Store in sealed containers and reuse.	Regulated as volatile organic compound emissions under the Clean Air Act.
Methanol (MeOH)	11,265.00	S-111, S-117	Liquid organic waste	Collect in sealed containers and reuse. Prevent release to water systems.	Regulated under the Clean Air Act for air emissions and Clean Water Act for wastewater discharge,
Water (H_2O)	3,792,046	S-151B, S-151C, S-151D, S-153	Process wastewater	Collect plant wastewater and treat before reuse or discharge.	Regulated under the Clean Water Act through National Pollutant Discharge Elimination System permits.

6.3 Material Safety

Material safety is a critical aspect of chemical process design in ensuring the safety of workers, equipment, the environment, and surrounding communities from hazardous substances and unsafe operating conditions. In this project, the conversion of captured carbon dioxide into olefins involves the handling of several flammable, toxic, and high-pressure materials including hydrogen, methanol, carbon monoxide, and volatile organic hydrocarbon products. Proper material safety management requires understanding of the physical and chemical properties of these substances, including flammability, toxicity, corrosiveness, reactivity, and exposure risks. Safety measures such as leak detection systems, pressure relief devices, proper ventilation, emergency shutdown procedures, and the use of personal protective equipment (PPE) are essential to minimize operational hazards. In addition, strict compliance with environmental and industrial safety regulations ensures that all materials are stored, transported, and processed safely throughout the facility. By implementing comprehensive material safety practices, the process can maintain reliable operation while reducing risks to personnel and the environment.

Carbon Dioxide (CO_2):

Carbon dioxide produced in the process must be managed due to its classification as greenhouse gas. Emissions are typically released through controlled vent systems and monitored to ensure compliance with regulatory requirements. Facilities emitting large quantities of CO_2 must report emissions under the EPA Greenhouse Gas Reporting Program (40 CFR Part 98) when annual emissions exceed 25,000 metric tons. Management strategies focus on minimizing emissions through efficient process operation and potential capture technologies. Carbon dioxide emissions are regulated under the Clean Air Act climate regulations (Greenhouse Gas Reporting Program (40 CFR Part 98) n.d.) (U.S. Environmental Protection Agency (EPA) n.d.)

Methanol (MeOH):

Methanol can pose environmental hazards when released in high concentrations due to its toxicity to aquatic organisms. Environmental management practices focus on preventing releases through closed systems, proper storage, and vapor recovery where possible. In the event of a spill or release, quantities greater than the CERCLA reportable quantity of 5,000 lb. must be reported to regulatory authorities. Wastewater containing methanol must be treated prior to discharge to ensure compliance with the (U.S. Environmental Protection Agency (EPA) n.d.)

Water (H_2O):

Water itself does not present a direct environmental hazard; however, process water may contain contaminants such as dissolved organics or reaction byproducts. Environmental management therefore focuses on the treatment of wastewater prior to discharge. Industrial wastewater must be treated to meet pollutant concentration limits, temperature limits, and chemical oxygen demand standards before being discharged to municipal treatment systems or surface waters under Clean Water Act NPDES permits (Resource Conservation and Recovery Act Hazardous Waste Regulations n.d.) (U.S. Environmental Protection Agency (EPA) n.d.)

Carbon Monoxide (CO):

Carbon monoxide is classified as an air pollutant and must be controlled to prevent adverse effects on air quality and human health. Environmental management typically involves routing carbon monoxide–containing streams to combustion or oxidation systems where CO is converted into carbon dioxide. Emissions must comply with air quality limits established under the Clean Air Act National Ambient Air Quality Standards (NAAQS), and facilities are required to obtain appropriate air emission permits (Greenhouse Gas Reporting Program (40 CFR Part 98) n.d.) (U.S. Environmental Protection Agency (EPA) n.d.)

Hydrogen (H_2):

Hydrogen is a highly flammable gas and is managed primarily through safe handling and storage procedures. Environmental concerns are generally associated with accidental releases rather than long-term environmental toxicity. Facilities storing large quantities of hydrogen must comply with OSHA Process Safety Management (29 CFR 1910.119) requirements, which regulate the handling of highly hazardous chemicals and require safety management systems to prevent accidental releases (National Institute for Occupational Safety and Health (NIOSH). n.d.) (U.S. Environmental Protection Agency (EPA) n.d.)

Ethylene (C_2H_4):

Ethylene emissions are regulated because they contribute to atmospheric pollution and photochemical reactions in the atmosphere. Environmental management focuses on minimizing emissions through closed processing systems and recovery or recycling of the gas where possible. Ethylene emissions are regulated under the Clean Air Act volatile organic compound emission standards and are subject to facility air emission permits (Resource Conservation and Recovery Act Hazardous Waste Regulations n.d.) (U.S. Environmental Protection Agency (EPA) n.d.)

Propylene (C_3H_6):

Propylene is classified as a volatile organic compound and can contribute to smog formation when released into the atmosphere. Environmental management involves minimizing emissions through leak detection, vapor recovery systems, and controlled combustion when disposal is required. Propylene emissions are regulated under EPA volatile organic compound emission regulations established under the Clean Air Act (Resource Conservation and Recovery Act Hazardous Waste Regulations n.d.) (U.S. Environmental Protection Agency (EPA) n.d.)

Butene (C_4H_8)

Butene is also classified as a volatile organic compound and must be managed to prevent atmospheric emissions. Facilities typically control emissions through vapor recovery systems, closed processing equipment, and combustion in flare systems when necessary. Environmental management of butene emissions must comply with Clean Air Act volatile organic compound standards and facility air emission permitting requirements (EPA, 2023).

Pentane (C_5H_{10}):

Pentene emissions are subject to air quality regulations due to their classification as volatile organic compounds. Environmental management practices focus on minimizing emissions through recovery, recycling, and controlled combustion when necessary. Facilities must also comply with Clean Air Act emission reporting and air quality regulations for hydrocarbon emissions (Greenhouse Gas Reporting Program (40 CFR Part 98) n.d.) (U.S. Environmental Protection Agency (EPA) n.d.)

Methane (CH_4):

Methane is a potent greenhouse gas with significant global warming potential. Environmental management strategies focus on minimizing methane releases by capturing and using methane as a fuel when possible or routing it to flare systems for combustion. Methane emissions are regulated under the EPA Greenhouse Gas Reporting Program, which requires facilities emitting significant quantities of methane to report emissions annually (Greenhouse Gas Reporting Program (40 CFR Part 98) n.d.) (U.S. Environmental Protection Agency (EPA) n.d.)

Methyl diethanolamine (MDEA):

Methyl diethanolamine used in gas treatment systems can present environmental hazards at high concentrations, particularly in wastewater streams. Environmental management involves proper containment, spill prevention, and treatment of wastewater before discharge. Wastewater containing MDEA must comply with pollutant limits established under the Clean Water Act and NPDES wastewater discharge permits to prevent contamination of groundwater or surface water (U.S. Environmental Protection Agency (EPA) n.d.) (Resource Conservation and Recovery Act Hazardous Waste Regulations n.d.)

SAPO-34 Catalyst (Silicoaluminophosphate):

SAPO-34 is a solid catalyst used in methanol-to-olefins processes. While it has relatively low environmental toxicity, catalyst dust can cause respiratory irritation and must be handled carefully. Environmental management focuses on proper handling, storage, and disposal of spent catalyst materials. Spent SAPO-34 catalysts are typically managed as industrial solid waste and must comply with disposal requirements under the Resource Conservation and Recovery Act (RCRA) (U.S. Environmental Protection Agency (EPA) n.d.) (Resource Conservation and Recovery Act Hazardous Waste Regulations n.d.)

$Cu/ZnO/Al_2O_3$ Catalyst:

The $Cu/ZnO/Al_2O_3$ catalyst is commonly used in methanol synthesis reactors to promote the hydrogenation of carbon monoxide and carbon dioxide into methanol. While the catalyst itself generally has low environmental mobility because it is a solid material, environmental management focuses on preventing the release of catalyst dust and properly handling spent catalyst material. Exposure to catalyst dust can pose health risks due to the presence of copper and zinc compounds, which may cause respiratory irritation if inhaled. For this reason, catalyst handling and replacement operations are typically conducted in controlled environments with appropriate containment and dust control measures.

Spent $Cu/ZnO/Al_2O_3$ catalyst must be properly characterized before disposal because it may contain accumulated heavy metals or hydrocarbons from the process. Environmental management typically involves collecting the spent catalyst in sealed containers and sending it to catalyst regeneration or metal recovery facilities when possible. If recycling is not feasible, the catalyst must be disposed of according to waste management regulations under the Resource Conservation and Recovery Act (RCRA), which governs the disposal of industrial solid and hazardous wastes. Facilities must ensure that catalyst waste is stored, transported, and disposed of in accordance with EPA waste management requirements to prevent contamination of soil or groundwater (U.S. Environmental Protection Agency (EPA) n.d.) (Resource Conservation and Recovery Act Hazardous Waste Regulations n.d.)

Table 6: Summary of Material Safety Data

Components	Boiling Point	Flash Point	Autoignition	Flammability Limits	Shelf-life	NFPA Health	NFPA Flammability	NFPA Reactivity
Carbon Dioxide (CO_2)	−109.3 °F (sublimes)	N/A	N/A	Nonflammable	Indefinite	3	0	0
Methanol (MeOH)	147 °F	52 °F	824 °F	Lower: 6% / Upper: 36%	2 months	1	3	0
Water (H_2O)	212 °F	N/A	N/A	Nonflammable	Indefinite	0	0	0
Carbon Monoxide (CO)	−312.6 °F	N/A	1128 °F	Lower: 12.5% / Upper: 74%	Indefinite	3	4	0
Hydrogen (H_2)	−423 °F	N/A	932 °F	Lower: 4% / Upper: 75%	Indefinite	0	4	0
Ethylene (C_2H_4)	−155 °F	−211 °F	842 °F	Lower: 2.7% / Upper: 36%	Indefinite	1	4	0
Propylene (C_3H_6)	−53.9 °F	−162 °F	842 °F	Lower: 2% / Upper: 11%	Indefinite	1	4	0
Butene (C_4H_8)	~20 °F	−80 °F	739 °F	Lower: 1.6% / Upper: 9.7%	Indefinite	1	4	0
Pentene (C_5H_{10})	~86 °F	−40 °F	617 °F	Lower: 1.5% / Upper: 7.8%	Indefinite	1	3	0
Methane (CH_4)	−258.7 °F	N/A	999 °F	Lower: 5% / Upper: 15%	Indefinite	0	4	0

Methyl diethanolamine (MDEA)	~477 °F	~260 °F	~509 °F	Combustible liquid	>1 year	2	1	0
SAPO-34 (Silicoaluminophosphate Catalyst)	N/A (solid)	N/A	N/A	Nonflammable solid	Indefinite, if stored dry	1	0	0
Copper/Zinc Oxide/Aluminum Oxide ($Cu/ZnO/Al_2O_3$ Catalyst)	N/A (solid)	N/A	N/A	Nonflammable solid	Indefinite, if stored dry	1	0	0

6.4 Operational Safety

Operational safety is essential to maintaining productive and accident-free operating conditions in chemical processing facilities. In the proposed CO_2 to olefins process, multiple high-temperature, high-pressure, and flammable operations are integrated throughout hydrogen production, methanol synthesis, methanol-to-olefins conversion, and olefin separation systems. These operations contain potential hazards such as fires, explosions, pressure buildup, equipment failure, and toxic gas releases if process conditions are not carefully controlled. To prevent and reduce these risks, the plant incorporates operational safety measures including automated process control systems, emergency shutdown procedures, pressure relief devices, continuous monitoring instrumentation, and preventive maintenance programs. Proper operator training and adherence to OSHA Process Safety Management (PSM) standards are also critical for ensuring safe plant operation. By implementing strong operational safety practices, the facility can maintain reliable production, protect personnel and equipment, and reduce the likelihood of environmental or industrial incidents.

6.4.1 Methanol Synthesis

Table 7: CO_2 TO Methanol Operational Safety Table

Process	**Unit**	**Safety Concerns**	**Prevention and Mitigation**

CO_2 **To Methanol**	CO_2 Hydrogenation Reactor	High Pressure Operation	• Reactor operates at elevated pressure due to hydrogen feed • Install pressure relief valves and rupture disks • Use continuous monitoring to prevent over-pressurization (OSHA 29 CFR 1910.119)
	CO_2 Hydrogenation Reactor	High Temperature / Hot Surfaces	• Catalytic hydrogenation occurs at high temperatures • Wear heat-resistant gloves, protective clothing, and face protection (OSHA 29 CFR 1910 Subpart I) • Insulate equipment to reduce burn hazards
	CO_2 Hydrogenation Reactor	Hydrogen Flammability	• Highly flammable; forms explosive mixtures with air • Control ignition sources and ground equipment • Use hydrogen leak detection and proper ventilation per OSHA standards
	Reactor / Product Stream	Methanol Toxicity and Flammability	• Methanol is toxic and highly flammable • OSHA exposure limits: 200 ppm TWA, 250 ppm STEL • Use ventilation, grounded storage, and PPE (gloves, goggles, flame-resistant clothing)

	Gas Handling System	CO_2 Exposure / Asphyxiation	• CO_2 can displace oxygen in confined spaces • OSHA PEL: 5000 ppm (8-hr TWA) • Use gas monitoring and proper ventilation to prevent oxygen deficiency

6.4.2 Methanol to Olefins

Table 8: Methanol to Olefins Operational Safety Table			
Process	**Unit**	**Safety Concerns**	**Prevention and Mitigation**
Methanol To Olefins	MTO Reactor	High Temperature / Hot Surfaces	• MTO reaction occurs at temperatures above 400 °C • Wear heat-resistant gloves, protective clothing, and face protection (OSHA 29 CFR 1910 Subpart I) • Use reactor insulation and temperature monitoring to prevent burns and overheating
	MTO Reactor	High-Pressure Operation	• Pressurized operation creates risk of over-pressurization • Install pressure relief valves and rupture disks • Use continuous monitoring systems (OSHA 29 CFR 1910.119)

	MTO Reactor / Catalyst System	Catalyst Handling Hazard (SAPO-34)	• SAPO-34 catalyst may produce dust during handling or replacement • Follow OSHA Hazard Communication Standard (29 CFR 1910.1200) and maintain SDS access • Wear respirators, gloves, and eye protection to prevent inhalation or skin exposure
	Reactor / Process Stream	Flammable Hydrocarbon Formation	• Produces flammable hydrocarbons (ethylene, propylene) • Control ignition sources and ground equipment (OSHA 29 CFR 1910.106) • Use proper ventilation and gas detection systems
	Product Gas Handling	Explosion / Gas Leak Hazard	• Olefin gases are highly flammable and can form explosive mixtures with air • Implement continuous leak detection and proper ventilation • Use emergency shutdown systems per OSHA Process Safety Management (29 CFR 1910.119)
	Methanol Feed System	Methanol Toxicity and Flammability	• Methanol is toxic and flammable • OSHA exposure limits: 200 ppm TWA, 250 ppm STEL • Use ventilation, closed transfer systems, and PPE (gloves, goggles,

			flame-resistant clothing)

6.4.3 Olefin Separation Units

Table 9: Olefin Separation Units Operational Safety Table			
<u>Process</u>	**<u>Unit</u>**	**<u>Safety Concerns</u>**	**<u>Prevention and Mitigation</u>**
<u>Olefin Separation Units</u>	Olefin Separation System	High Pressure Operation	• Separation units (distillation columns, compressors) operate at elevated pressure • Install pressure relief valves and rupture disks • Use continuous pressure monitoring (OSHA 29 CFR 1910.119)
	Separation Columns	Flammable Hydrocarbon Vapors	• Olefins (ethylene, propylene, light hydrocarbons) are highly flammable • Control ignition sources and ground equipment (OSHA 29 CFR 1910.106) • Install vapor leak detection systems

	Compression / Gas Handling Equipment	Explosion or Gas Leak Hazard	• Pressurized hydrocarbon gases can form explosive mixtures with air • Use proper ventilation and continuous gas monitoring • Implement emergency shutdown systems (OSHA 29 CFR 1910.119)
	Product Handling System	Cryogenic / Low Temperature Hazard	• Some olefin separations operate at very low temperatures • Risk of cold burns and equipment embrittlement • Wear insulated gloves and face protection (OSHA 29 CFR 1910 Subpart I)
	Hydrocarbon Product Streams	Worker Exposure to Hydrocarbon Vapors	• Hydrocarbon vapors may cause dizziness or respiratory irritation • Use proper ventilation, sealed transfer systems, and monitoring equipment • Maintain safe conditions per OSHA industrial hygiene guidelines

6.5 Hazard and Operability Study (HAZOP)

A Hazard and Operability Study (HAZOP) is a systematic risk assessment method used to identify potential process hazards, operational deviations, and safety concerns within the process. In the proposed CO_2 to olefins facility, the process involves complex operations including hydrogen production, methanol synthesis, high-pressure gas handling, catalytic reactions, and cryogenic separation systems, all of which present potential risks if abnormal operating conditions occur. The purpose of the HAZOP analysis is to evaluate how deviations in key process variables such as pressure, temperature, flow, and composition could impact equipment performance, product quality, personnel safety, and environmental protection. By using structured guide words and process reviews, HAZOP identifies possible causes, consequences, safeguards, and recommendations for each operating unit. This analysis supports the development of safer process designs, improved control strategies, and emergency response procedures while ensuring compliance with industrial safety standards and regulations

Table 11: R-101 (HAZOP) Analysis

Component	**Guide Word**	**Deviation**	**Possible Causes**	**Consequences**	**Existing Safeguards**	**Recommendations**
R-101	Less	Level	Reduced feed from S-107, blockage in inlet line, feed valve partially closed	Insufficient reactant inventory, unstable reaction, reduced conversion	Level indicator, feed flow monitoring on S-107	Install low-level alarm and inspect feed line S-107
	More	Level	Excess feed flow from S-107, outlet restriction in S-108, control valve failure	Reactor flooding, possible pressure buildup, carryover to downstream units	Level indicator, pressure monitoring	Install high-level alarm and automatic feed control
	More	Pressure	Gas buildup, blocked outlet line S-108, excessive	Reactor rupture, safety hazard	Pressure indicator, automatic pressure relief valve	Install high-pressure alarm and emergency relief system

			feed rate from S-107			
	Less	Pressure	Leak in reactor system, insufficient feed pressure from S-107	Reduced reaction efficiency	Pressure monitoring	Inspect reactor and piping for leaks
	More	Temperature	Excessive heat generation from reaction, cooling system failure	Thermal runaway, equipment damage, safety risk	Temperature sensors and cooling system	Install high-temperature trip and emergency cooling
	Less	Temperature	Insufficient heating, cold feed stream from S-107	Reduced reaction rate and poor conversion	Temperature monitoring	Improve heating control
	Reverse	Flow Direction	Pressure imbalance between reactor outlet S-108 and inlet S-107	Backflow into feed lines, contamination of upstream units	Check valves on feed lines	Maintain non-return valves
	No	Flow	Feed stream S-107 stopped, upstream valve closed,	Reactor starved of reactants, reaction stops, production loss	Flow indicator on S-107, control system alarms	Install low-flow alarm and interlock to stop reactor operation

			blockage in feed line			
	Less	Flow	Reduced feed from S-107, partial blockage in feed line, pump/control valve malfunction	Lower reactor throughput, reduced conversion and production rate	Flow monitoring on S-107	Inspect feed pipeline and maintain valves
	More	Flow	Excess feed from S-107, control valve failure open	Reduced residence time, incomplete reaction, unstable operation	Flow control loop on feed stream	Install high-flow alarm and automatic flow control

Table 10: S-107 (HAZOP) Analysis

Stream	**Guide Word**	**Deviation**	**Possible Causes**	**Consequences**	**Existing Safeguards**	**Recommendations**
S-107	No	Flow	Blockage in pipeline, valve closed, upstream interruption in S-106A, heat exchanger	Reactor R-101 receives no feed, reaction stops, production loss	Flow indicator, control system alarms	Install low-flow alarm and shutdown interlock

			E-101 failure			
	Less	Flow	Reduced flow from S-106A, fouling in E-101, partially closed valve	Reduced feed to R-101, lower conversion and production rate	Flow monitoring	Inspect heat exchanger and inlet piping
	More	Flow	Control valve failure open, excessive upstream flow from S-106A	Reactor R-101 overloaded, reduced residence time and incomplete reaction	Flow control loop	Install high-flow alarm and automatic flow control
	More	Pressure	Downstream restriction in R-101, valve closed at reactor inlet	Pressure buildup in feed line, potential pipeline damage	Pressure indicator	Install pressure relief device or pressure alarm
	Less	Pressure	Leak in pipeline, pressure drop from S-106A	Unstable feed conditions entering R-101, reduced reaction efficiency	Pressure monitoring	Inspect pipeline and connections for leaks

	More	Temperature	Excess heating in E-101, high upstream temperature from S-106A	Possible overheating in R-101, increased risk of unsafe reaction conditions	Temperature indicator	Install high-temperature alarm
	Less	Temperature	Insufficient heating in E-101, low upstream temperature	Reduced reaction rate and poor conversion in R-101	Temperature monitoring	Improve heat exchanger temperature control
	Reverse	Flow Direction	Higher pressure in reactor outlet S-108 than feed line S-107	Backflow toward E-101 or upstream piping, contamination of upstream units	Check valves	Maintain non-return valves and monitor pressure balance

Table 11: S-108 (HAZOP) Analysis

Stream	**Guide Word**	**Deviation**	**Possible Causes**	**Consequences**	**Existing Safeguards**	**Recommendations**
S-108	No	Flow	Blockage in outlet pipeline, valve closed, reactor R-	Reactor pressure buildup, possible reactor	Pressure indicator, reactor control system	Install high-pressure alarm and emergency relief system

			101 shutdown	damage or shutdown		
	Less	Flow	Partial blockage in outlet line, fouling, downstream blockage in S-108A/S-108B	Increased residence time in R101, possible overheating	Flow monitoring	Inspect outlet piping and downstream lines
	More	Flow	Excess feed from S-107, control valve failure open	Reduced residence time in R-101, incomplete reaction	Flow monitoring, reactor control system	Install high-flow alarm and improve feed control
	More	Pressure	Downstream blockage in S-108A or S-108B, closed valve	Pressure buildup in reactor R-101, potential safety hazard	Pressure indicator, pressure relief valve	Install high-pressure shutdown system
	Less	Pressure	Leak in pipeline, pressure drop in downstream lines	Reduced flow stability and inefficient downstream processing	Pressure monitoring	Inspect pipeline and fittings for leaks
	More	Temperature	High temperature from reactor	Thermal stress on downstream equipment	Temperature indicator	Install high-temperature alarm

			reaction conditions			
	Less	Temperature	Rapid cooling in pipeline, heat loss to surroundings	Reduced efficiency in downstream separation or processing	Temperature monitoring	Add insulation to outlet piping
	Reverse	Flow Direction	Higher pressure in downstream streams S-108A/S-108B than reactor outlet	Backflow toward reactor R-101, disturbance of reactor operation	Check valves and flow direction	Maintain non-return valves and monitor pressure balance

6.6 Personal Protective Equipment

Personal Protective Equipment (PPE) is a crucial component of industrial safety that helps protect workers from exposure to physical, chemical, thermal, and respiratory hazards present in chemical processing facilities. In the proposed CO_2 to olefins plant, personnel may encounter hazardous conditions associated with high-pressure systems, elevated temperatures, cryogenic temperatures flammable gases, toxic chemicals such as methanol and carbon monoxide, and moving industrial equipment. To minimize the risk of injury or exposure, workers must use appropriate PPE including flame resistant clothing, chemical-resistant gloves, safety goggles, hearing protection, and respiratory protection when necessary. The selection and proper use of PPE must comply with OSHA safety standards and plant specific safety protocols to ensure adequate protection during routine operations, maintenance activities, and emergency situations. Combined with proper training and safe operating procedures, PPE serves as the final barrier of protection between personnel and workplace hazards, helping maintain a safe and reliable operating environment.

Table 12: Personal Protective Equipment Table

PPE needed for plant safety	Application for a safe working area.
Flame-Resistant (FR) Clothing	Required when working around flammable gases and hydrocarbons such as hydrogen, ethylene, and propylene in the hydrogenation reactor, MTO reactor, and olefin separation units. Protects personnel from flash fire hazards.
Chemical-Resistant Jumpsuits	Used during MDEA handling, catalyst loading, and chemical maintenance activities to prevent skin contact with corrosive or irritating chemicals and contaminated process streams.
Heat-Resistant Gloves and Sleeves	Required when working near high-temperature equipment such as the CO_2 hydrogenation reactor and MTO reactor which operate at elevated temperatures. Protects against burns from hot surfaces.
Chemical-Resistant Aprons or Suits	Used during amine solution transfer, spill cleanup, or chemical maintenance in the CO_2 capture system to reduce exposure to MDEA and other chemical solutions.
Respirators / Cartridge Respirators	Required during catalyst handling (SAPO-34 dust), maintenance operations, or areas with potential vapor exposure to prevent inhalation of particulate matter or hazardous vapors.

Cryogenic-Rated Protective Gloves and Face Shield	Required during low-temperature olefin separation processes where cryogenic conditions may cause cold burns or frostbite.
Hard Hat with Face Shield Attachment	Used during maintenance operations, catalyst loading/unloading, and pressurized equipment servicing where mechanical or impact hazards may occur.

7 Economic Analysis

The economic evaluation of the integrated process indicates a capital-intensive design with strong long-term profitability driven by high-value olefin products and carbon incentives. The total capital investment for the plant is estimated at $740.2 million, with the majority attributed to Inside Battery Limits (ISBL) costs at $423.0 million, reflecting the complexity of the reaction, separation, and purification units required for MTO and upstream processing. Outside Battery Limits (OSBL) contribute an additional $169.2 million, covering utilities, storage, and supporting infrastructure. Contingency and engineering costs are each estimated at $42.3 million, while working capital accounts for $63.4 million, ensuring sufficient liquidity for plant startup and operation. Fixed capital cost is reported as $46.9 million, completing the overall investment structure.

The ISBL cost distribution highlights that the hydrogen production and CO_2 capture section dominates capital expenditure, accounting for $228 million, which is more than half of the ISBL costs. This is consistent with the energy-intensive nature of steam methane reforming and high-pressure absorption systems. The CO_2-to-methanol synthesis section contributes $110 million, reflecting the cost of high-pressure catalytic reactors and separation systems, while the MTO section contributes $86 million, indicating a comparatively lower but still significant investment in fluidized-bed reactors and downstream processing.

Further breakdown of ISBL costs by unit operation shows that distillation columns represent the largest equipment cost at approximately $54 million, emphasizing the importance and complexity of cryogenic separation in achieving polymer-grade olefins. Furnaces and heat exchangers also contribute significantly, at approximately $34 million and $12 million, respectively, due to the high thermal demands of reforming and reaction sections. Other equipment such as compressors, absorbers, flash drums, and pumps contribute smaller but non-negligible portions, collectively reflecting the highly integrated nature of the process. The presence of multiple separation steps and recycle loops further increases equipment requirements.

The variable cost of production (VCOP) is driven primarily by utilities, which account for approximately $145 million annually, making it the largest operating expense. This is expected given the high energy requirements of SMR, compression, and cryogenic distillation. The hydrogen production and CO_2 capture section contributes $78 million, while CO_2 feed costs contribute $15 million and MDEA solvent costs contribute $21 million, reflecting the ongoing expenses associated with chemical absorption and regeneration. Catalyst costs are relatively minor, with both $Cu/ZnO/Al_2O_3$ and SAPO-34 catalysts contributing approximately $0.4 million each, indicating that catalyst replacement is not a dominant economic factor compared to utilities and raw materials.

Fixed costs of production are largely influenced by labor, maintenance, and overhead. The plant operates with three shifts per day and six operators per shift, resulting in an annual operator salary of $1.08 million, with additional supervision and overhead bringing total labor-related costs higher. Maintenance costs are estimated at 3% of ISBL, or $20.30 million annually, while overhead expenses, including taxes, insurance, and land, contribute $24.62 million per year. These values are consistent with large-scale chemical processing facilities and reflect the complexity and scale of the operation.

Revenue analysis demonstrates strong economic potential, with total annual revenue estimated at $424.37 million. The largest contributor is propylene at $168.63 million, followed by ethylene at $75.91 million, and the C_4 stream at $56.45 million. A significant portion of revenue is also derived from 45Q tax incentives, contributing $123.37 million annually, highlighting the importance of carbon capture integration in improving economic feasibility. The 45V incentive contributes a negligible amount, indicating that hydrogen-related incentives are less impactful in this design. Overall, the process achieves a gross profit of $309.62 million, demonstrating strong profitability driven by both product sales and policy incentives.

The net present value (NPV) analysis shows that the project experiences negative cash flow during the initial years, corresponding to capital investment and startup costs, followed by a transition to positive cash flow as revenue generation begins. The NPV becomes positive around year 12, indicating a moderate payback period consistent with large-scale petrochemical investments. Beyond this point, the NPV increases steadily, reaching a substantial positive value by the end of the project lifetime, confirming long-term economic viability.

Sensitivity analysis further highlights the economic drivers of the process. Variations in revenue have the most significant impact on NPV, demonstrating the project's dependence on product pricing and market conditions. Changes in VCOP and ISBL also influence profitability but to a lesser extent, while fixed costs have a relatively minor effect. The comparison between MACRS-5 and MACRS-7 depreciation schedules shows similar trends, with slightly faster cost recovery under MACRS-5 leading to improved early cash flow, though both scenarios ultimately yield positive long-term returns.

Overall, the economic evaluation demonstrates that while the process requires substantial upfront investment and is heavily influenced by energy costs, it achieves strong profitability through high-value olefin production and carbon credit incentives. The integration of CO_2 capture not only enhances environmental performance but also plays a critical role in improving the financial feasibility of the plant.

Table 13: Capital Cost Summary ($MM)	
ISBL	423.0
OSBL	169.2
Contingency	42.3
Engineering Cost	42.3
Working Capital	63.4
Fixed Capital Cost	46.9
Total Capital Costs	740.2

Figure 7 shows the ISBL cost split across the major unit operations in the process. The largest cost contribution comes from hydrogen production and CO_2 capture, at approximately $228 million, indicating that upstream feed preparation and carbon capture are the most capital-intensive sections of the plant. This is expected because these units require major equipment such as reactors, compressors, absorbers, heat exchangers, and separation systems to generate hydrogen and purify CO_2 for downstream conversion. The CO_2-to-methanol section represents the second-largest cost at about $110 million, reflecting the expense of catalytic reactors, recycle systems, and separation equipment needed to convert CO_2 and hydrogen into methanol. The methanol-to-olefins section contributes the smallest share, at approximately $86 million, but remains a significant portion of the ISBL due to the specialized MTO reactor system, catalyst handling, and product recovery requirements. Overall, the figure highlights that the economic performance of the process is heavily influenced by the cost of hydrogen generation and CO_2 capture, making these areas important targets for cost reduction and process optimization.

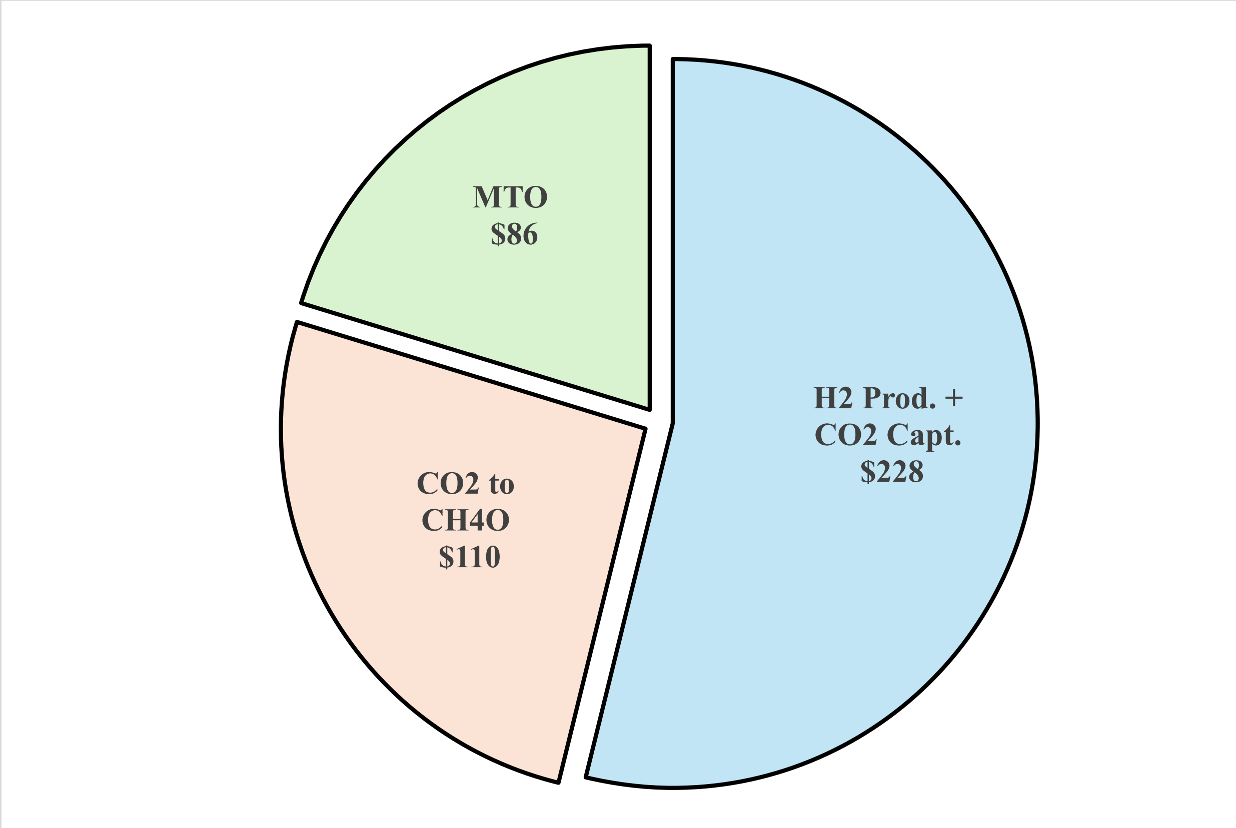

Figure 7: ISBL Split into Major Unit Operations

Figure 8 presents the ISBL cost split by equipment type, showing which categories of equipment contribute most strongly to the total inside battery limits capital cost. The largest cost contribution comes from distillation columns, at approximately $54 million, indicating that separation and purification are major cost drivers in the process. This is expected because the process requires multiple separation steps to purify methanol, remove water and unreacted gases, and recover polymer-grade ethylene and propylene products.

The next major equipment costs are associated with compressors, at approximately $34 million, and heat exchangers, at about $12 million. Compressors are significant because the process involves high-pressure CO_2 hydrogenation, gas recycle, and downstream product recovery, all of which require pressure adjustments. Heat exchangers also play an important role because temperature control is necessary across reactors, condensers, reboilers, and cooling steps.

Smaller cost contributions come from equipment such as reaction vessels, dryers, furnaces, pumps, absorbers, knockout drums, flash drums, splitters, and water treatment units. Although each of these individual equipment types has a smaller cost compared to the major separation and compression equipment, they are still essential for maintaining process performance, product purity, and safe operation. Overall, the figure shows that the ISBL cost is dominated by separation and compression equipment, meaning that future cost optimization should focus on reducing the number, size, or energy demand of columns and compressors.

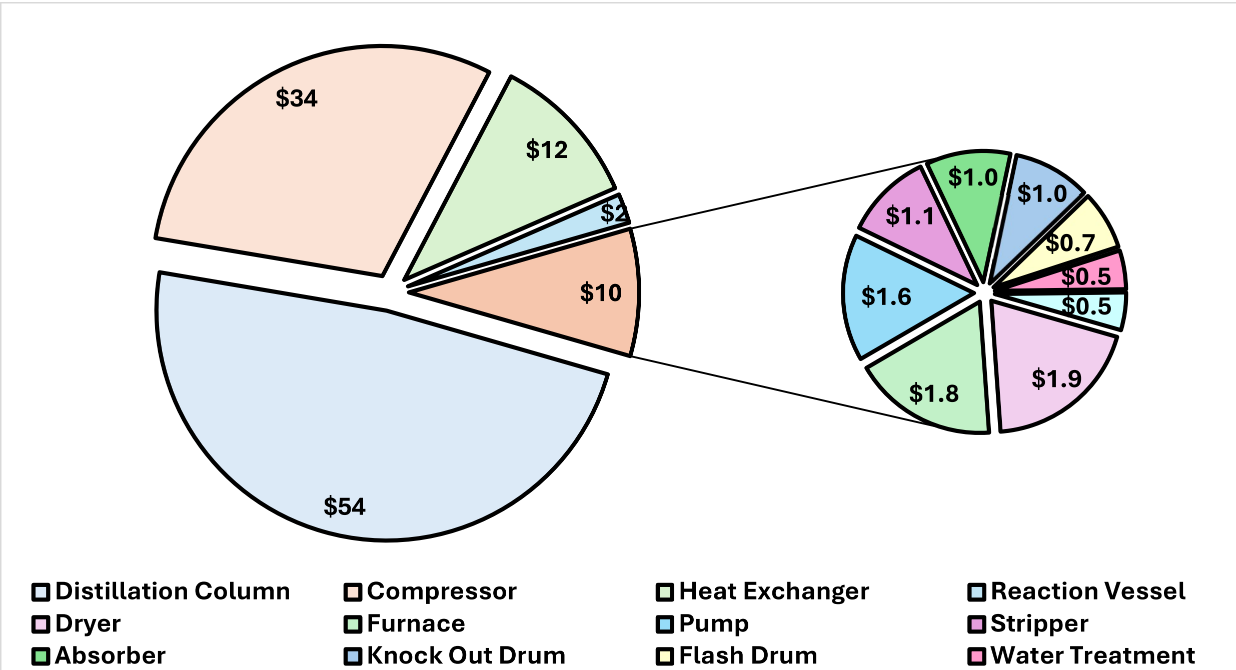

Figure 8: ISBL Cost Split into Equipment Type

Figure 9 shows the VCOP cost contributors for the process, highlighting the major variable costs required for plant operation. The largest contributor is utilities, at approximately \$145 million, making it the dominant operating expense. This reflects the high energy demand of the process, including heating, cooling, compression, distillation, and cryogenic separation requirements.

The second-largest contribution comes from hydrogen production and CO_2 capture, at approximately \$78 million. This cost is significant because producing or supplying hydrogen and capturing CO_2 require energy-intensive equipment and supporting materials. Since hydrogen is a major reactant in the CO_2-to-methanol section, this cost strongly influences the overall economics of the process.

Raw material and solvent-related costs are smaller but still important. MDEA contributes approximately \$21 million, reflecting its use in the CO_2 capture system, while CO_2 contributes about \$15 million as a feedstock cost. Catalyst costs are much smaller in comparison, with both $Cu/ZnO/Al_2O_3$ and SAPO-34 contributing about \$0.4 million each. Although these catalyst costs are relatively low compared to utilities and hydrogen-related costs, they remain essential because they directly support the CO_2-to-methanol and methanol-to-olefins reactions.

Overall, the figure indicates that the process economics are most sensitive to utility demand, hydrogen cost, and CO_2 capture expenses. Therefore, improving heat integration, reducing compression requirements, optimizing CO_2 capture, and securing low-cost hydrogen would have the greatest impact on lowering the variable cost of production

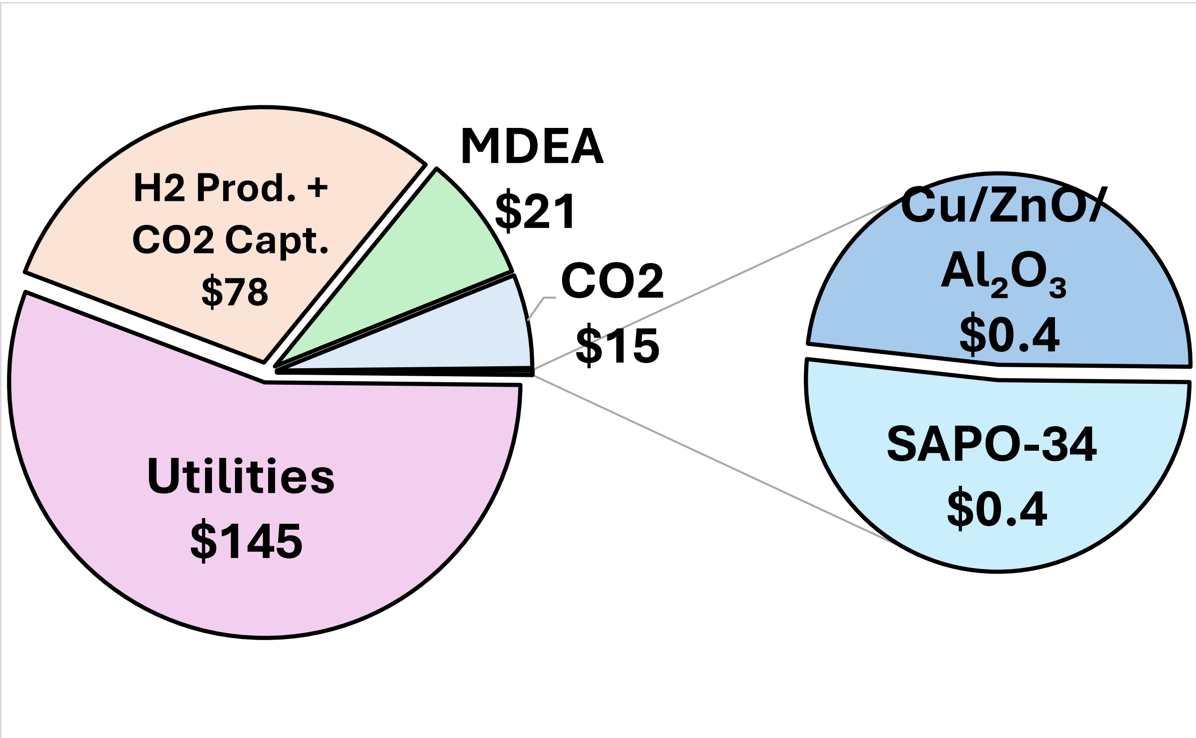

Figure 9: VCOP Cost Contributors

Table 14 summarizes the fixed cost of production for the proposed process. The plant is assumed to operate with 3 shifts per day and 6 operators per shift, which supports continuous operation and provides the labor coverage needed for a large-scale chemical production facility. The total operator salary cost is estimated at approximately $1.08 million per year, while supervision costs are calculated as 25% of operator labor, contributing an additional $0.27 million per year. Direct salary overhead, which includes labor-related expenses such as benefits and administrative support, is estimated at 45% of the combined labor and supervision cost, adding $0.61 million per year.

The largest fixed cost contributors are maintenance and overhead expenses. Maintenance is estimated as 3% of the ISBL cost, resulting in a yearly cost of approximately $20.30 million. This high value reflects the scale and complexity of the process equipment, including compressors, reactors, distillation columns, heat exchangers, and CO_2 capture units. Overhead expenses, which include plant overhead, taxes, insurance, and land-related costs, contribute approximately $24.62 million per year, making it the largest fixed cost category in the table.

Overall, the fixed cost of production is dominated by maintenance and overhead expenses, while labor-related costs represent a smaller portion of the total. This indicates that the process is capital-intensive, meaning the long-term economics depend heavily on equipment reliability, plant uptime, and effective maintenance planning.

Table 14: Fixed Cost of Production	
Number of Shifts per day	3
Number of Operators per Shift	6
Operator Salary ($/yr)	1.08
Supervision (25% of Operator) ($/yr)	0.27
Direct Salary Overhead (45% of Labor and Supervision.) ($/yr)	0.61
Maintenance (3% ISBL) ($/yr)	20.30
Overhead Expenses (Plant, Taxes, Insurance, Land) ($/yr)	24.62

Table 15 summarizes the revenue analysis for the proposed CO_2-to-olefins process. The main revenue sources come from the sale of ethylene, propylene, and the C4 stream, along with federal tax incentives from 45Q carbon capture credits and 45V hydrogen-related credits. The total annual revenue is estimated to be approximately $424.37 million per year, while the gross profit is estimated at about $309.62 million per year.

Among the chemical products, propylene provides the largest product revenue, generating approximately $168.63 million per year from an annual production rate of 168,630 tons/year at a selling price of $1,000/ton. Ethylene contributes approximately $75.91 million per year, based on a production rate of 108,448 tons/year and a selling price of $700/ton. The C4 stream also provides a meaningful revenue contribution of about $56.45 million per year, showing that byproduct recovery improves the overall economic performance of the process rather than treating the C4 fraction as waste.

The 45Q tax incentive is also a major contributor to revenue, adding approximately $123.37 million per year based on 1,542,102 tons/year of CO_2 at a credit value of $80/ton. This makes the tax credit one of the most important economic drivers for the project, nearly comparable to the revenue from propylene sales. In contrast, the 45V tax incentive contributes a very small amount, approximately $0.00017 million per year, and therefore has a negligible effect on the overall revenue compared to product sales and the 45Q credit.

Overall, the revenue analysis shows that the process is economically supported by both chemical product sales and carbon capture incentives. Propylene sales and the 45Q tax credit are the dominant contributors, while ethylene and the C4 stream provide additional revenue that strengthens the profitability of the process. The estimated gross profit of $309.62 million per year suggests that the process has strong economic potential, especially if product pricing remains favorable and the project qualifies for the expected carbon capture tax incentives.

Table 15: Revenue Analysis

Product	Price ($/ton)	Rate (ton/yr)	Price ($MM/yr)
Ethylene (Intratec n.d.)	700	108,448	75.91
Propylene (Propylene Market Size, Share & Industry Analysis 2026)	1000	168,630	168.63
C4 Stream (Petroleum & Other Liquids Prices 2026)	600	94,091	56.45
45Q Tax Incentive (Jones 2023)	80	1,542,102	123.37
45v Tax Incentive (10-yr) (Jones 2023)	0.003	56,445	0.00017
		Total Revenue	424.37
		Gross Profit	309.62

The Net Present Value figure shows the projected cumulative economic performance of the process over the full project lifetime. During the first several years, the NPV remains negative because the project requires a large upfront capital investment during construction and early operation. The most negative values occur near the beginning of the project, reflecting the impact of equipment purchasing, installation, and other startup-related costs before the plant reaches full revenue generation.

After the initial investment period, the NPV gradually improves as the plant begins producing and selling ethylene, propylene, and the C4 stream while also receiving revenue support from the 45Q tax incentive. The trend becomes less negative each year, showing that the annual cash flows are strong enough to recover the initial investment over time. The project reaches its break-even point around year 12, where the NPV crosses from negative to positive. This indicates that the discounted cash inflows have recovered the original investment and operating costs by that point.

From year 13 onward, the NPV continues to increase steadily, ending at approximately $350 million by year 22. This positive final NPV suggests that the project is economically attractive under the assumed product prices, production rates, tax incentives, and operating costs. Overall, the figure shows that although the process is capital-intensive and requires a long payback period, it has strong long-term profitability once full operation is achieved.

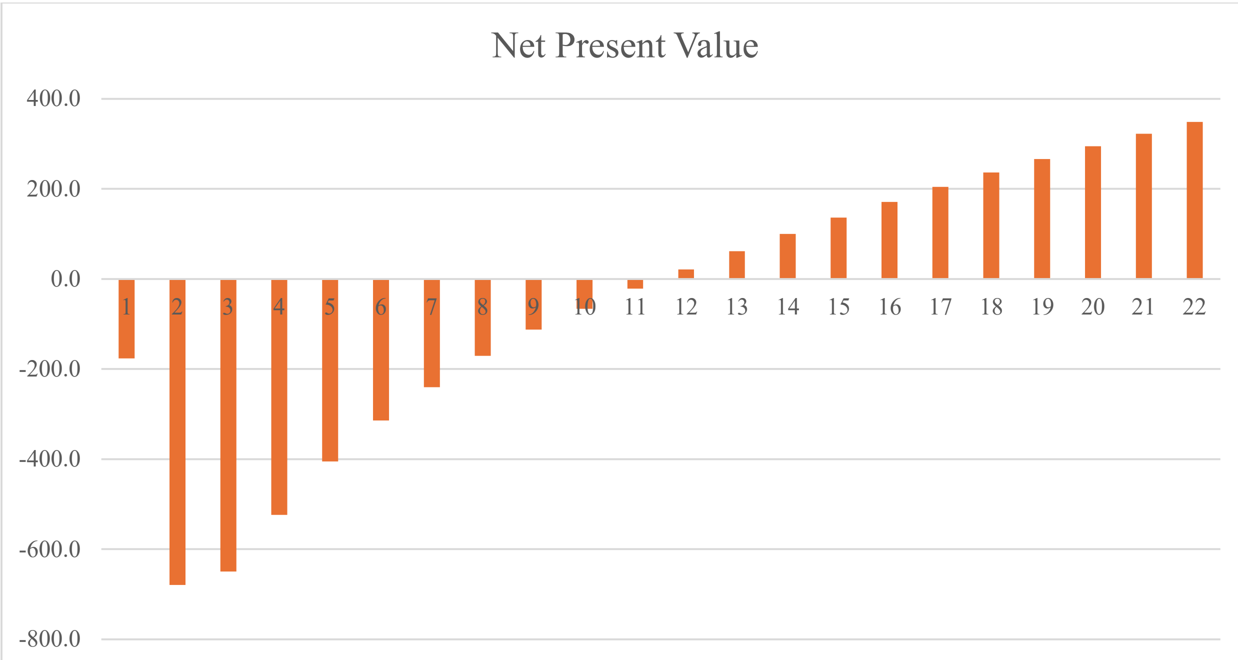

Figure 10: Net Present Value Analysis

This sensitivity analysis shows how changes in major economic variables affect the overall project profitability. The x-axis represents a ±30% change in each variable, while the y-axis shows the resulting change in project value. All four variables intersect at 0% change, which represents the base-case economic scenario.

The most sensitive variable is revenue, shown by the steep upward trend. When revenue decreases by 30%, the project value drops significantly, while a 30% increase in revenue greatly improves profitability. This indicates that the process economics are highly dependent on product selling prices, production rates, and tax incentives such as 45Q. Because revenue has the steepest slope, small changes in market pricing or incentive eligibility have the strongest effect on the project's financial outcome.

The next most important variable is VCOP, or variable cost of production. As VCOP increases, project value decreases sharply, showing that operating costs such as utilities, hydrogen production, CO_2 capture, MDEA, and feedstock costs strongly influence profitability. A decrease in VCOP improves the economic performance, while an increase reduces the project's financial attractiveness.

The ISBL cost and FCOP have smaller slopes compared to revenue and VCOP, meaning they have a less severe impact on profitability. ISBL cost still affects the economics because it represents major capital equipment expenses, such as reactors, compressors, distillation columns, and separation units. FCOP has the least dramatic effect, suggesting that fixed costs such as labor, maintenance, supervision, and plant overhead are less influential than revenue and variable operating costs.

Overall, the graph shows that the project is most sensitive to revenue and variable operating costs, while capital and fixed costs have a smaller impact. This means the project's economic success depends most heavily on maintaining strong product revenues, qualifying for tax incentives, reducing utility demand, and controlling hydrogen and CO_2 capture costs.

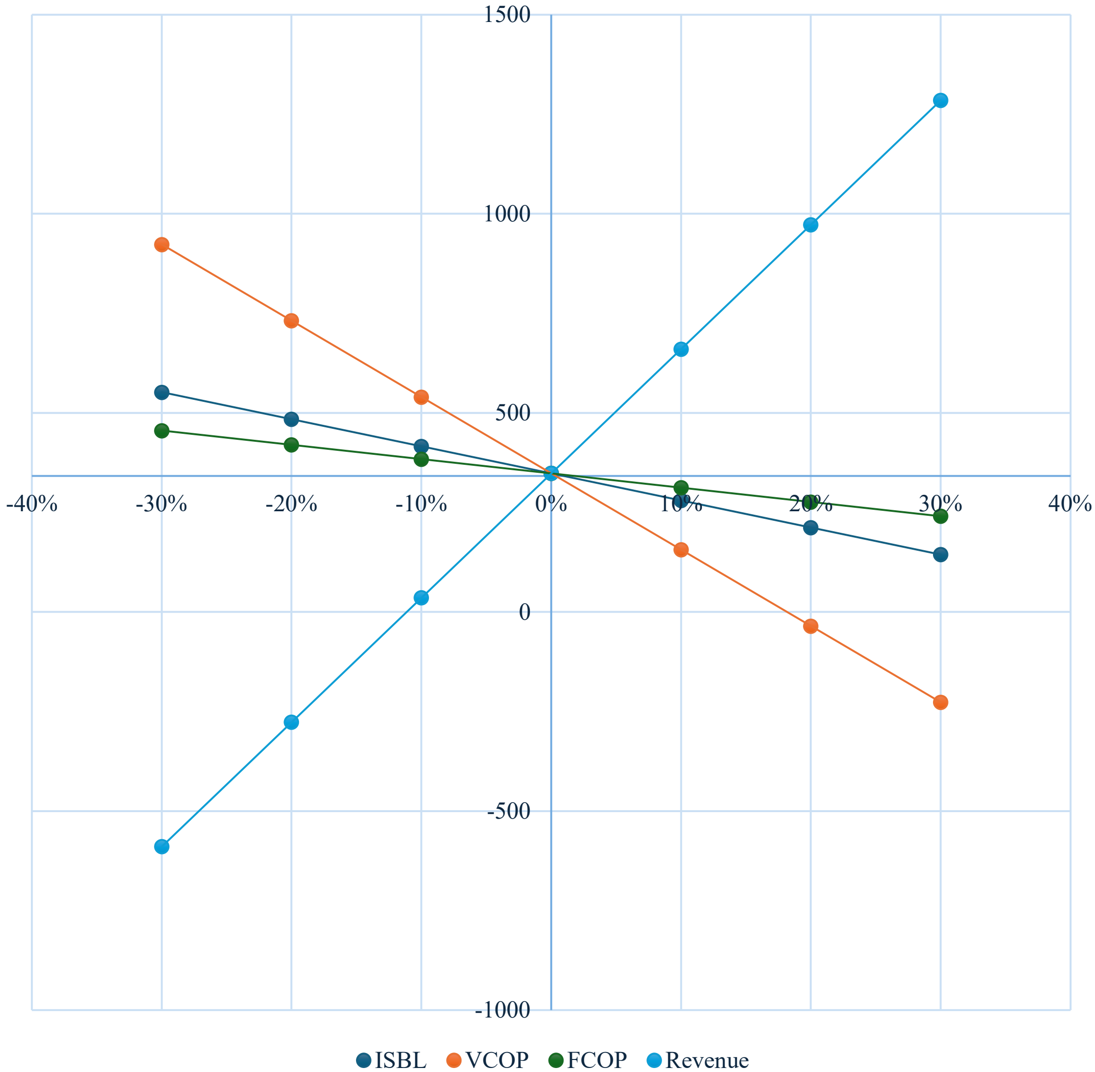

Figure 11: Sensitivity Analysis with MACRS-5

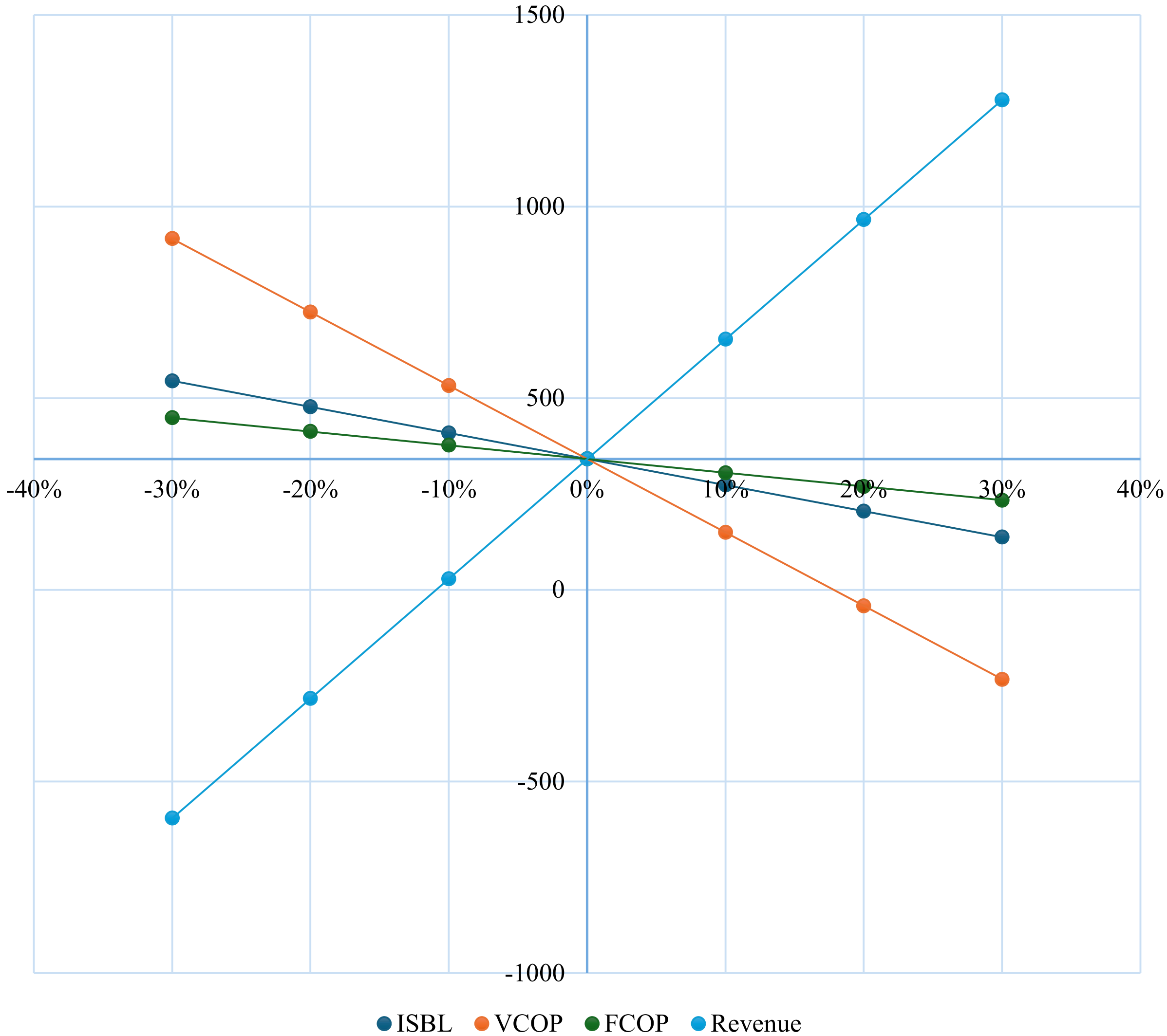

Figure 12: Sensitivity Analysis with MACRS-7

8 Conclusion

This project demonstrates that converting captured carbon dioxide into light olefins is both a technically feasible and economically promising pathway toward sustainable chemical production. By integrating carbon capture, hydrogen production, methanol synthesis, and methanol to olefins (MTO) conversion into a single process, the design successfully transforms CO_2 from a waste emission into a valuable feedstock. The system achieves high conversion efficiencies, produces polymer-grade ethylene and propylene at an industrial scale (80,000 tons/year), and incorporates recycling strategies that significantly reduce net emissions. From an economic perspective, the process shows strong profitability potential, supported by substantial annual revenues, and enhanced further by policy incentives such as 45Q carbon credits. Although capital costs are high, primarily driven by hydrogen production and CO_2

capture, the project remains financially viable and capable of reaching breakeven within a typical timeframe for large-scale chemical facilities.

Environmentally, the design represents a significant improvement over conventional fossil-based olefin production. The integration of carbon capture, CO_2 recycling, and energy efficient heat integration results in substantial reductions in greenhouse gas emissions, aligning the process with global decarbonization goals and regulatory requirements. Additionally, the ability to produce low-carbon olefins provides a competitive advantage in a market increasingly driven by sustainability and ESG commitments.

Looking forward, future objectives exist to further enhance the viability and scalability of this process. Development should focus on implementing direct and verifiable CO_2 capture methods to fully qualify for tax credits and reduce reliance on pipeline feedstocks. Process optimization efforts should aim to improve overall efficiency while reducing water consumption, hydrogen demand, and electricity usage, thereby lowering both operating costs and environmental impact. Expanding the business model to include licensing or as a service, both are approaches similar to industry leaders like Honeywell UOP. That could provide resilience in scenarios where Carbon Capture tax incentives are removed. Additionally, integrating downstream polymer production would significantly increase product value and profitability, while expansion into international markets would enable broader adoption of this technology and strengthen its global impact.

Overall, the conceptual plant concludes that CO_2 to olefins technology represents a compelling pathway toward a circular carbon economy. With continued innovation, optimization, and strategic expansion, this process has strong potential for commercialization and long-term transformation of the chemical industry toward a more sustainable and economically resilient future.

9 Appendix

9.1 Appendix I: Design Basis

The objective of this design is to utilize post-combustion CO_2 emissions from existing pipelines as a sustainable carbon feedstock to produce polymer-grade ethylene (99.9 mol%) and propylene (99.5 mol%). The process integrates post-combustion CO_2 capture using a molecular sieve system, followed by hydrogen supply and conditioning for downstream reaction requirements. Captured CO_2 is converted to methanol through catalytic hydrogenation, and the methanol intermediate is subsequently transformed into light olefins via a methanol-to-olefins (MTO) reactor employing a SAPO-34 catalyst. The resulting olefin products are then purified to meet polymer-grade specifications. The facility is designed to operate continuously at large scale, with emphasis on maximizing CO_2 utilization, optimizing energy efficiency, ensuring economic feasibility, and minimizing overall environmental impact.

The proposed facility is designed for large-scale olefin production to meet industrial demand while achieving significant CO_2 reuse.

Table 16: Plant Size Specifications	
Parameter	**Value**
Annual Operating Time	8000 h/y
Construction Period	2 years
Start-up Capacity	50% design capacity during 3rd year
Plant Life	20 years
Methanol Production (intermediate)	275,710 lb/hr
Ethylene Production	28,540 lb/hr
Propylene Production	47,400 lb/hr

The selected plant location is Houston, Texas, situated within the U.S. Gulf Coast refining and petrochemical corridor. This site was chosen due to its proximity to large refinery flue gas sources, ensuring a consistent and reliable supply of CO_2 for capture and conversion. The region also provides access to established hydrogen infrastructure and industrial utilities necessary to support methanol synthesis and MTO operations. Additionally, Houston's close proximity to polymerization facilities enables direct sales of polymer-grade ethylene and propylene, reducing transportation costs and logistical complexity. The area is supported by extensive port, pipeline, and rail infrastructure, facilitating efficient distribution of products and raw materials. Furthermore, federal incentives promoting carbon capture and utilization (CCUS) strengthen the economic viability of the project. Overall, Houston's highly integrated petrochemical ecosystem enhances supply chain efficiency and improves the overall feasibility of producing polymer-grade olefins from captured CO_2.

9.2 Appendix II: Block Flow Diagram

The block flow diagram of the process depicts the fully defined processes in white and the black-boxed processes in black.

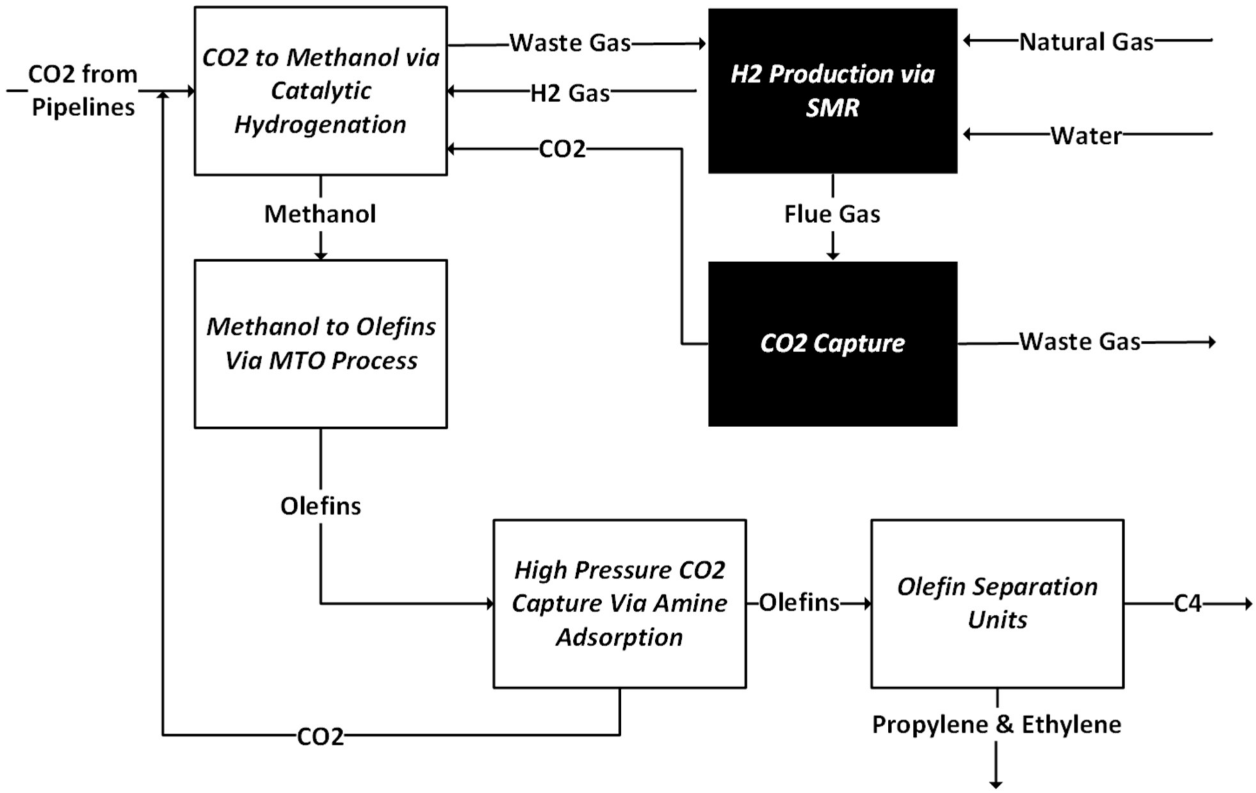

Figure 13: Block Flow Diagram of the Entire Process

9.3 Appendix III: Process Flow Diagram and Material Balance

	CO_2-S101	H_2-S103	H_2O-S121	H_2O-S122	MOH-S120	MOH-S123	PUR-S111	S128G
Phase	**Vapor**	**Vapor**	**Liquid**	**Liquid**	**Liquid**	**Vapor**	**Vapor**	**Vapor**
Temp (°F)	95	104	464	464	384	464	95	397
Press (psia)	1,098	363	566	566	566	566	1,065	1,098
Mass Flows (lb/hr)	348,805	124,439	164,541	151,122	275,710	13,419	117,686	38,490
Mass Flow (lb/hr)								
CO_2	348,805	0	0	0	999	0	36,159	30,539
H_2	0	124,439	0	0	0	0	61,142	0
MeOH	0	0	13,419	0	274,423	13,419	2,574	0
CO	0	0	0	0	0	0	11,332	0
H_2O	0	0	151,122	151,122	288	0	267	0
O_2	0	0	0	0	0.104	0	829	969
N2	0	0	0	0	0	0	5,382	6,982

	S140	S105	S106	S106A	S106B	S107	S108	S108A
Phase	**Vapor**	**Vapor**	**Vapor**	**Vapor**	**Vapor**	**Vapor**	**Vapor**	**Vapor**
Temp (°F)	140	316	147	239	333	410	506	413
Press (psia)	580	566	566	566	566	1,105	1,096	1,096
Mass Flows (lb/hr)	23,519	535,252	1,594,425	1,594,425	1,594,425	1,568,263	1,568,263	1,568,263
Mass Flow (lb/hr)								
CO_2	958	380,301	705,736	705,736	705,736	684,970	314,205	314,205
H_2	0	124,439	674,717	674,717	674,717	675,887	622,246	622,246
MeOH	0	0	23,163	23,163	23,163	23,096	314,435	314,435
CO	22,561	22,561	124,553	124,553	124,553	117,894	99,190	99,190
H_2O	0	0	2,403	2,403	2,403	2,399	154,171	154,171
O_2	0	969	8,434	8,434	8,434	8,596	8,596	8,596
N2	0	6,982	55,420	55,420	55,420	55,420	55,420	55,420

	S108C	S109	S110	S112	S113	S114	S115	S116A
Phase	**Vapor**	**Vapor + Liquid**	**Vapor**	**Vapor**	**Vapor**	**Liquid**	**Vapor + Liquid**	**Vapor + Liquid**
Temp (°F)	411	95	95	95	102	95	98	59
Press (psia)	1,096	1,044	1,065	1,065	1,098	1,065	17	88
Mass Flows (lb/hr)	1,568,263	1,618,065	1,176,859	1,059,173	1,059,173	441,207	441,207	441,207
Mass Flow (lb/hr)								
CO_2	314,205	363,472	361,594	325,435	325,435	1,878	1,878	1,878
H_2	622,246	611,440	611,420	550,278	550,278	20	20	20
MeOH	314,435	313,618	25,736	23,163	23,163	287,882	287,882	287,882
CO	99,190	113,330	113,325	101,992	101,992	6	6	6
H_2O	154,171	154,083	2,670	2,403	2,403	151,414	151,414	151,414
O_2	8,596	8,296	8,294	7,464	7,464	3	3	3
N2	55,420	53,825	53,820	48,438	48,438	5	5	5

	S117	S118	S118A	S118B	S119	S151C	S151D	S117
Phase	**Vapor**	**Liquid**	**Liquid**	**Liquid**	**Liquid**	**Liquid**	**Liquid**	**Vapor**

Temp (°F)	59	59	68	69	176	447	248	59
Press (psia)	17	17	17	580	580	435	435	17
Mass Flows (lb/hr)	955	440,251	440,251	440,251	440,251	979,285	979,285	955
Mass Flow (lb/hr)								
CO_2	879	999	999	999	999	0	0	879
H_2	20	0	0	0	0	0	0	20
MeOH	40	287,842	287,842	287,842	287,842	0	0	40
CO	6	0	0	0	0	0	0	6
H_2O	4	151,410	151,410	151,410	151,410	979,285	979,285	4
O_2	2	0	0	0	0	0	0	2
N2	5	0	0	0	0	0	0	5

	C3-S159	**C4-S161**	**C5-S162**	**CH4-S168**	**ETH-S171**	**ETL-S170**	**H_2O-S122**	**H_2O-S137**
Phase	**Liquid**	**Liquid**	**Liquid**	**Vapor**	**Liquid**	**Liquid**	**Liquid**	**Liquid**
Temp (°F)	123	124	197	14	33	-6	464	122
Press (psia)	261	73	73	464	363	363	566	580
Mass Flows (lb/hr)	925	26,451	359	9,261	140	28,540	151,122	149,398
Mass Flow (lb/hr)								
CO_2	0	0	0	0	0	0	0	0
H_2	0	0	0	4,305	0	0	0	0
MeOH	0	0	0	0	0	0	0	0
CO	0	0	0	0	0	0	0	0
C_2H_4	0	0	0	2,247	4	28,535	0	0
H_2O	0	0	0	0	0	0	151,122	149,398
C_3H_6	80	11	0	0	0	0	0	0
C_2H_6	0	0	0	0	136	4	0	0
C_3H_8	844	390	0	0	0	0	0	0
C_4H_8	0	26,049	4	0	0	0	0	0
C_5H_{10}	0	0	355	0	0	0	0	0
CH_4	0	0	0	2,709	0	0	0	0
MDEA	0	0	0	0	0	0	0	0
C	0	0	0	0	0	0	0	0
O_2	0	0	0	0	0	0	0	0
N_2	0	0	0	0	0	0	0	0

	H_2O-S151	MOH-S123	PG-S151D	PL-S157	MOH-S121	S124	S125	S126
Phase	Liquid	Vapor	Liquid	Liquid	Liquid	Liquid	Vapor + Liquid	Vapor
Temp (°F)	144	464	248	118	384	385	385	752
Press (psia)	429	566	435	290	566	580	566	580
Mass Flows (lb/hr)	77	13,419	293,786	47,401	275,710	275,710	293,004	293,004
Mass Flow (lb/hr)								
CO_2		0	0	0	999	999	959	959
H_2		0	0	0	0	0	0	0
MeOH		13,419	0	0	274,423	274,423	291,751	291,751
CO		0	0	0	0	0	0	0
C_2H_4		0	0	3	0	0	3	3
H_2O		0	293,786	0	288	288	288	288
C_3H_6		0	0	47,278	0	0	2	2
C_2H_6		0	0	114	0	0	0	0
C_3H_8		0	0	6	0	0	0	0
C_4H_8		0	0	0	0	0	0	0
C_5H_{10}		0	0	0	0	0	0	0
CH_4		0	0	0	0	0	0	0
MDEA		0	0	0	0	0	0	0
C		0	0	0	0	0	0	0
O_2		0	0	0	0	0	0	0
N_2		0	0	0	0	0	0	0

	S127	S128	S128A	S128C	S128D	S128E	S128F	S128G
Phase	Vapor + Liquid	Solid Phase	Vapor	Vapor	Vapor	Vapor + Liquid	Vapor + Liquid	Vapor
Temp (°F)	752	752	752	124	111	111	140	140
Press (psia)	580	580	580	566	566	566	566	566
Mass Flows (lb/hr)	293,004	8,335	311,727	311,727	273,237	61,172	61,172	38,490
Mass Flow (lb/hr)								
CO_2	959	0	30,539	30,539	0	30,539	30,539	30,539
H_2	4,239	0	0	0	0	0	0	0
MeOH	26	0	0	0	0	0	0	0
CO	22,561	0	0	0	0	0	0	0
C_2H_4	30,315	0	0	0	0	0	0	0
H_2O	149,796	0	0	0	0	2,979	2,979	0

C_3H_6	46,635	0	0	0	0	0	0	0
C_2H_6	251	0	0	0	0	0	0	0
C_3H_8	1,222	0	0	0	0	0	0	0
C_4H_8	25,649	0	0	0	0	0	0	0
C_5H_{10}	350	0	0	0	0	0	0	0
CH_4	2,667	0	0	0	0	0	0	0
MDEA	0	0	0	0	0	19,703	19,703	0
C	8,335	8,335	0	0	0	0	0	0
O_2	0	0	48,461	48,461	47,492	969	969	969
N_2	0	0	232,727	232,727	225,745	6,982	6,982	6,982

	S128H	S128I	S128J	S129	S129B	S131	S132	S133
Phase	Liquid	Liquid	Liquid	Vapor	Vapor	Vapor	Vapor + Liquid	Vapor + Liquid
Temp (°F)	554	554	509	77	446	752	296	122
Press (psia)	580	580	580	16	580	580	580	566
Mass Flows (lb/hr)	22,682	22,682	22,682	303,392	311,727	284,670	284,670	284,670
Mass Flow (lb/hr)								
CO_2	0	0	0	0	30,539	959	959	959
H_2	0	0	0	0	0	4,239	4,239	4,239
MeOH	0	0	0	0	0	26	26	26
CO	0	0	0	0	0	22,561	22,561	22,561
C_2H_4	0	0	0	0	0	30,315	30,315	30,315
H_2O	2,979	2,979	2,979	0	0	149,796	149,796	149,796
C_3H_6	0	0	0	0	0	46,635	46,635	46,635
C_2H_6	0	0	0	0	0	251	251	251
C_3H_8	0	0	0	0	0	1,222	1,222	1,222
C_4H_8	0	0	0	0	0	25,649	25,649	25,649
C_5H_{10}	0	0	0	0	0	350	350	350
CH_4	0	0	0	0	0	2,667	2,667	2,667
MDEA	19,703	19,703	19,703	0	0	0	0	0
C	0	0	0	0	0	0	0	0
O_2	0	0	0	70,665	48,461	0	0	0
N_2	0	0	0	232,727	232,727	0	0	0

	S134	S135	S136	S138	S139	S140	S141	S142
Phase	Vapor	Liquid	Vapor + Liquid	Vapor + Liquid	Vapor + Liquid	Vapor	Liquid	Liquid

Temp (°F)	122	122	122	127	140	140	162	162
Press (psia)	580	580	580	580	580	580	580	580
Mass Flows (lb/hr)	135,239	149,430	32	43,236	43,236	23,519	18,406	18,406
Mass Flow (lb/hr)								
CO_2	958	1	1	958	958	958	0	0
H_2	4,238	0	0	0	0	0	0	0
MeOH	1	25	25	13	13	0	16	16
CO	22,561	0	0	22,561	22,561	22,561	0	0
C_2H_4	30,311	3	3	0	0	0	0	0
H_2O	398	149,398	0	0	0	0	0	0
C_3H_6	46,634	2	2	0	0	0	0	0
C_2H_6	251	0	0	0	0	0	0	0
C_3H_8	1,222	0	0	0	0	0	0	0
C_4H_8	25,648	0	0	0	0	0	0	0
C_5H_{10}	350	0	0	0	0	0	0	0
CH_4	2,667	0	0	0	0	0	0	0
MDEA	0	0	0	19,703	19,703	0	18,390	18,390
C	0	0	0	0	0	0	0	0
O_2	0	0	0	1	1	0	1	1
N_2	0	0	0	0	0	0	0	0

	S143	**S144**	**S145**	**S146**	**S147**	**S148**	**S149**	**S150**
Phase	**Liquid**	**Liquid**	**Liquid**	**Liquid**	**Vapor**	**Vapor**	**Vapor**	**Vapor**
Temp (°F)	140	104	136	86	124	210	165	147
Press (psia)	580	580	580	580	284	508	508	508
Mass Flows (lb/hr)	18,406	1,512	19,914	19,914	111,918	111,918	111,918	111,918
Mass Flow (lb/hr)								
CO_2	0	0	0	0	0	0	0	0
H_2	0	0	0	0	4,238	4,238	4,238	4,238
MeOH	16	0	12	12	0	0	0	0
CO	0	0	0	0	0	0	0	0
C_2H_4	0	0	0	0	30,311	30,311	30,311	30,311
H_2O	0	199	199	199	596	596	596	596
C_3H_6	0	0	0	0	46,634	46,634	46,634	46,634
C_2H_6	0	0	0	0	251	251	251	251
C_3H_8	0	0	0	0	1,222	1,222	1,222	1,222
C_4H_8	0	0	0	0	25,648	25,648	25,648	25,648
C_5H_{10}	0	0	0	0	350	350	350	350

CH_4	0	0	0	0	2,667	2,667	2,667	2,667
MDEA	18,390	1,314	19,703	19,703	0	0	0	0
C	0	0	0	0	0	0	0	0
O_2	1	0	1	1	0	0	0	0
N_2	0	0	0	0	0	0	0	0

	S151A	S151B	S151C	S151D	S151E	S152	S153	S154
Phase	Liquid	Liquid	Liquid	Liquid	Liquid	Vapor	Liquid	Vapor
Temp (°F)	265	428	447	248	248	144	143	143
Press (psia)	435	435	435	435	435	446	446	446
Mass Flows (lb/hr)	979,285	979,285	979,285	979,285	685,500	113,680	603	113,077
Mass Flow (lb/hr)								
CO_2	0	0	0	0	0	0	0	0
H_2	0	0	0	0	0	4,305	0	4,305
MeOH	0	0	0	0	0	0	0	0
CO	0	0	0	0	0	0	0	0
C_2H_4	0	0	0	0	0	30,789	0	30,789
H_2O	979,285	979,285	979,285	979,285	685,500	603	603	0
C_3H_6	0	0	0	0	0	47,369	0	47,369
C_2H_6	0	0	0	0	0	254	0	254
C_3H_8	0	0	0	0	0	1,241	0	1,241
C_4H_8	0	0	0	0	0	26,053	0	26,053
C_5H_{10}	0	0	0	0	0	355	0	355
CH_4	0	0	0	0	0	2,709	0	2,709
MDEA	0	0	0	0	0	0	0	0
C	0	0	0	0	0	0	0	0
O_2	0	0	0	0	0	0	0	0
N_2	0	0	0	0	0	0	0	0

	S155	S156	S158	S160	S163	S164	S165
Phase	Liquid	Vapor + Liquid	Liquid	Liquid	Vapor	Vapor	Vapor
Temp (°F)	149	149	233	229	-88	-45	36
Press (psia)	305	305	290	261	305	305	508
Mass Flows (lb/hr)	75,135	75,135	27,735	26,810	37,942	37,942	37,942
Mass Flow (lb/hr)							
CO_2	0	0	0	0	0	0	0

H_2	0	0	0	0	4,305	4,305	4,305
MeOH	0	0	0	0	0	0	0
CO	0	0	0	0	0	0	0
C_2H_4	3	3	0	0	30,786	30,786	30,786
H_2O	0	0	0	0	0	0	0
C_3H_6	47,369	47,369	92	11	0	0	0
C_2H_6	114	114	0	0	141	141	141
C_3H_8	1,241	1,241	1,235	390	0	0	0
C_4H_8	26,053	26,053	26,053	26,053	0	0	0
C_5H_{10}	355	355	355	355	0	0	0
CH_4	0	0	0	0	2,709	2,709	2,709
MDEA	0	0	0	0	0	0	0
C	0	0	0	0	0	0	0
O_2	0	0	0	0	0	0	0
N_2	0	0	0	0	0	0	0

	S166	S167	S169	S200	S201	S203
Phase	Vapor + Liquid	Vapor	Liquid	Liquid	Liquid	Liquid
Temp (°F)	-67	-161	12	104	483	86
Press (psia)	508	464	464	580	580	580
Mass Flows (lb/hr)	37,942	9,261	28,680	1,512	22,682	22,682
Mass Flow (lb/hr)						
CO_2	0	0	0	0	0	0
H_2	4,305	4,305	0	0	0	0
MeOH	0	0	0	0	0	0
CO	0	0	0	0	0	0
C_2H_4	30,786	2,247	28,540	0	0	0
H_2O	0	0	0	199	2,979	2,979
C_3H_6	0	0	0	0	0	0
C_2H_6	141	0	141	0	0	0
C_3H_8	0	0	0	0	0	0
C_4H_8	0	0	0	0	0	0
C_5H_{10}	0	0	0	0	0	0
CH_4	2,709	2,709	0	0	0	0
MDEA	0	0	0	1,314	19,703	19,703
C	0	0	0	0	0	0
O_2	0	0	0	0	0	0
N_2	0	0	0	0	0	0

9.4 Appendix IV: Sample Hand Calculations

This appendix presents the sample hand calculations used to verify the preliminary design and sizing of Distillation Column D-101 for the CO_2 to Methanol process. The calculations were performed as a validation step to compare theoretical design methods with the automated results generated by ASPEN+ simulation software. Standard chemical engineering design procedures, including the Fenske equation, McCabe–Thiele analysis, and tray efficiency correlations, were applied to estimate the minimum and actual number of trays, column height, and column diameter. These calculations provide a practical check on the simulation results and demonstrate the engineering assumptions and methodologies used in the distillation column design process.

Minimum Number of Stages:

The minimum number of theoretical stages was calculated using the Fenske equation:

$$N_{min} = \frac{\ln\left(\frac{z_D}{1-z_D} \cdot \frac{1-z_B}{z_B}\right)}{\ln(\alpha)}$$

$$N_{min} = \frac{\ln\left(\frac{0.99}{0.01} \cdot \frac{0.93}{0.07}\right)}{\ln(4)} = 5.18$$

Minimum number of stages = 6 trays

Using McCabe–Thiele analysis for theoretical stages: **12 trays**

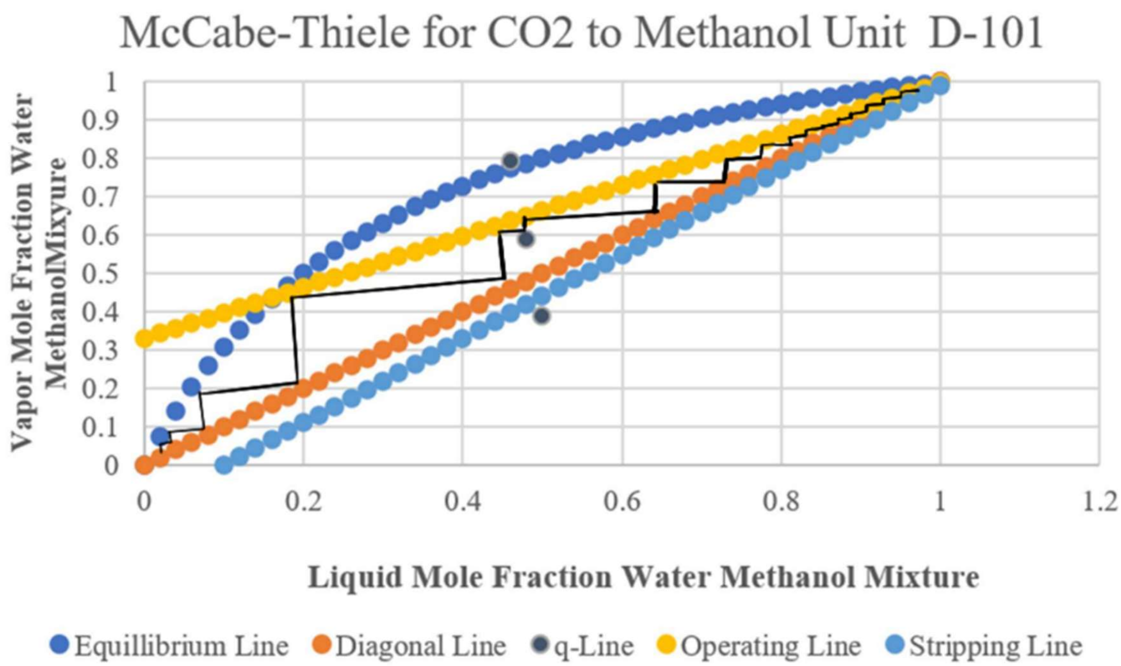

Figure 14: Absorber Section for Methanol to Olefins Process

Actual number of trays needed for Distillation Column (D-101):

$$E_0 = 0.70$$

$$N_{actual} = \frac{N_{min}}{E_0} = \frac{12}{0.70} = 17 \text{ trays}$$

<u>Column Height</u>

Tray spacing:

$$S = 2.0 \text{ ft}$$

$$H_{active} = N_{actual} \times S = 17 \times 2.0 = 34 \text{ ft}$$

Including reboiler and condenser allowances:

$$H_{total} = H_{active} + H_{reboiler} + H_{condenser}$$

$$H_{total} = 34 + 12 + 8 = 52.0 \text{ ft}$$

Total column height = 52.0 ft

<u>Column Diameter</u>

$$u_v = (-0.17L_t^2 + 0.27L_t - 0.047)\left(\frac{\rho_L - \rho_V}{\rho_V}\right)^{1/2}$$

$$u_v = (-0.17(0.61)^2 + 0.27(0.61) - 0.047)\left(\frac{790 - 8}{8}\right)^{1/2}$$

$$u_v = 0.83 \text{ m/s} = 2.71 \text{ ft/s}$$

Maximum vapor flowrate:

$$\dot{V} = (R + 1)D$$

$$\dot{V} = (2 + 1)(3531) = 10{,}593 \text{ kmol/hr}$$

<u>Volumetric Flowrate</u>

Using the ideal gas equation:

$$\dot{V} = \frac{nRT}{P}$$

$$\dot{V} = \frac{(10593)(8314)(353)}{1.23 \times 10^5}$$

$$\dot{V} = 70.2 \text{ m}^3/\text{hr} = 229.7 \text{ ft}^3/\text{s}$$

<u>Diameter Calculation</u>

$$D_c = \sqrt{\frac{4\dot{V}}{\pi u_v}}$$

$$D_c = \sqrt{\frac{4(70.2)}{\pi(0.96)}}$$

$$D_c = 9.63 \text{ m} = 31.6 \text{ ft}$$

Final Results

Table 17: Comparison of Hand Calculation and Aspen Results for D-101		
Parameter	**Hand Calculation**	**Aspen Results**
Column Trays	17 trays	15 trays
Column Diameter	31.6 ft	28.5 ft
Column Height	52.0 ft	50.2 ft

This appendix presents the sample hand calculations used to verify the preliminary design and sizing of Distillation Column D-101 for the CO_2 to Methanol process. The calculations were performed as a validation step to compare theoretical design methods with the automated results generated by ASPEN+ simulation software. Standard chemical engineering design procedures, including the Fenske equation, McCabe–Thiele analysis, and tray efficiency correlations, were applied to estimate the minimum and actual number of trays, column height, and column diameter. These calculations provide a practical check on the simulation results and demonstrate the engineering assumptions and methodologies used in the distillation column design process.

9.5 Appendix V: Utilities

Table 18: Cost of Utilities per Unit			
Equipment	**Utility Type**	**Utility Usage Usage**	**Cost ($/hr)**
C103	Electricity	1,369 kW	89.484
C102	Electricity	4,159 kW	268.452
D104	Cooling Water + HP Steam	6,465,617 kJ/hr	29.754
D103	Cooling Water + HP Steam	183,802,563 kJ/hr	868.611
E106	Cooling Water	553,027 kJ/hr	0.195

E103	Refrigerated water	18,185,402 kJ/hr	80.486
E104	HP Steam	49,742,669 kJ/hr	431.797
D105	Coolin Water + HP Steam	28,509,158 kJ/hr	126.172
E101	HP Steam	217,008,338 kJ/hr	2138.904
D107	Refrigerant	30,517,483 kJ/hr	148.712
C104	Electricity	608 kW	40.267
FH-101	Natural Gas	151,382,994 kJ/hr	908.160
C101	Electricity	37,594 kW	2237.1
D106	Refrigerant	306,011,142 kJ/hr	1313.833
P102	Electricity	11 kW	0.894
P103	Electricity	7 kW	0.134
E109	Cooling water	6,260,929 kJ/hr	2.213
P104	Electricity	8 kW	0.223
D101	Cooling Water + HP Steam	889,909,994 kJ/hr	5277.865
E107	Cooling Water	3,072,842 kJ/hr	1.086
E108	Cooling Water	26,912,288 kJ/hr	9.513
E105	Cooling Water	59,834,731 kJ/hr	21.151
P101	Electricity	413 kW	31.319
E102	Cooling Water	1,154,150,407 kJ/hr	407.993

9.6 Appendix VI: Equipment

This appendix includes the main equipment used in the process design and the basic sizing information for each unit. The table lists the equipment type, material of construction, estimated size, heat duty, and power requirements used in the design calculations. Material of construction was determined upon the heat and operating conditions. Carbon Steel was the basis for general application and did not require extreme conditions. Stainless steel is used during high temperature and high-pressure operations. Finally, Chromium lined reactors are used to prevent runoff reactions and corrosive damage to the facility.

9.6.1 Sizing

Table 19: Equipment List and Sizing

Equipment	Type	Material of Construction	Sizing	Duty (MMBtu/hr)	Work (hp)
C101	Centrifugal Compressor	Carbon Steel	—	—	50,401
C102	Centrifugal Compressor	Carbon Steel	—	—	5,579
C103	Centrifugal Compressor	Carbon Steel	—	—	1,836
C104	Centrifugal Compressor	Carbon Steel	—	—	816
P101	Centrifugal Pump	Stainless Steel	—	—	554
P102	Centrifugal Pump	Stainless Steel	—	—	15
P103	Centrifugal Pump	Stainless Steel	—	—	12
P103	Centrifugal Pump	Stainless Steel	—	—	10
R101	Packed Bed Reactor	Internal Chromium lining with Stainless Steel shell,	Diameter: 6.5 ft Length: 20 ft	0 (adiabatic)	—
R102	Fluidized Bed Reactor	Stainless Steel	Diameter: 12.2 ft Height: 36.6 ft	0 (adiabatic)	—
R103	Packed Bed Reactor	Stainless Steel	Diameter: 7.7 ft Height: 23.3 ft	0 (adiabatic)	—
D101	Tray Tower	Stainless Steel	Diameter: 16.5 ft Height: 134 ft	Reboiler: 482 Condenser: -361	—

D102	Tray Tower	Stainless Steel	Diameter: 7.5 ft Height: 98 ft	Reboiler: 16.70 Condenser: -32.29	—
D103	Tray Tower	Stainless Steel	Diameter: 13 ft Height: 92 ft	Reboiler: 87.41 Condenser: -86.80	—
D104	Tray Tower	Stainless Steel	Diameter: 25 ft Height: 92 ft	Reboiler: 2.976 Condenser: -3.151	—
D105	Tray Tower	Stainless Steel	Diameter: 6 ft Height: 92 ft	Reboiler: 12.56 Condenser: -14.45	—
D106	Tray Tower	Stainless Steel	Diameter: 15 ft Height: 92 ft	Reboiler: 142.8 Condenser: -147.2	—
D107	Tray Tower	Stainless Steel	Diameter: 5 ft Height: 342 ft	Reboiler: 14.21 Condenser: -14.71	—
ABS101	Absorber	Stainless Steel	Diameter: 4.5 ft Height: 12 ft	—	—
ABS102	Absorber	Stainless Steel	Diameter: 6 ft Height: 12 ft	—	—
STRIP101	Stripper	Stainless Steel	Diameter: 4 ft Height: 12 ft	—	—
STRIP102	Stripper	Stainless Steel	Diameter: 4 ft Height: 12.5 ft	—	—
KO101	Flash Drum	Stainless Steel	Diameter: 14 ft Height: 14.5 ft	0 (adiabatic)	—
FLT101	Flash Drum	Stainless Steel	Diameter: 8.5 ft Height: 25.5 ft	0 (adiabatic)	—
FLT102	Flash Drum	Stainless Steel	Diameter: 6 ft Height: 18.5 ft	0 (adiabatic)	—
FLT103	Flash Drum	Stainless Steel	Diameter: 4 ft	0 (adiabatic)	—

			Height: 12 ft		
WT101	Absorber	Stainless Steel	Diameter: 6.5 ft Height: 19.5 ft	—	—
WT102	Absorber	Stainless Steel	Diameter: 3 ft Height: 12 ft	—	—
E101	Steam Heater	Carbon Steel	Area: 23,168 sqft	206	—
E102	Cooling Heat Exchanger	Carbon Steel	Area: 46,656 sqft	-1093	—
E103	Cooling Heat Exchanger	Carbon Steel	Area: 5,458 sqft	-17.23	—
E104	Heating Heat Exchanger	Stainless Steel	Area: 2,127 sqft	47	—
E105	Cooling Heat Exchanger	Carbon Steel	Area: 3,797 sqft	-56.7	—
E106	Cooling Heat Exchanger	Carbon Steel	Area: 196 sqft	-0.5	—
E107	Cooling Heat Exchanger	Carbon Steel	Area: 200 sqft	-2.9	—
E108	Cooling Heat Exchanger	Carbon Steel	Area: 2,092 sqft	-25.5	—
E109	Cooling Heat Exchanger	Carbon Steel	Area: 494 sqft	-5.9	—

RE101	Shell and Tube Heat Exchanger	Carbon Steel	Area: 5,898 sqft	-235	—
RE102	Shell and Tube Heat Exchanger	Carbon Steel	Area: 39,154 sqft	242	—
RE103	Shell and Tube Heat Exchanger	Carbon Steel	Area: 74.45 sqft	3.9	—
RE104	Shell and Tube Heat Exchanger	Carbon Steel	Area: 30,785 sqft	198	—
RE105	Shell and Tube Heat Exchanger	Stainless Steel	Area: 1,612 sqft	24.5	—
RE106	Shell and Tube Heat Exchanger	Stainless Steel	Area: 13.5 sqft	0.8	—
RE107	Shell and Tube Heat Exchanger	Stainless Steel	Area: 82.5 sqft	0.2	—
RE108	Shell and Tube Heat Exchanger	Stainless Steel	Area: 34.9 sqft	1.2	—
RE109	Shell and Tube Heat Exchanger	Stainless Steel	Area: 340 sqft	3.0	—
FH101	Fire Heater	Stainless Steel	Area: 31,186 sqft	143	—

9.6.2 Pricing

Table 20: Equipment Price

Equipment	Installed Cost [USD]
ABS-101	174,100

ABS-102	209,600
C101	15,693,300
C102	8,232,600
C103	1,739,500
C103	1,561,900
D101	14,453,200
D102	166,900
D103	2,115,300
D104	897,600
D105	1,312,500
D106	4,278,200
D107	11,217,100
DR101	2,238,600
E101	1,438,000
E102	3,564,500
E103	281,400
E104	229,100
E105	319,500
E106	102,500
E107	105,800
E108	238,700
E109	140,600
FH-101	1,781,386
FLT101	308,000
FLT102	234,000
FLT103	158,100
KO101	1,012,200
P101	416,000
P102	120,600
P103	46,900
P104	53,100
R101	32,726
R102	1,985,400
R103	311,743
RE101	658,200
RE102	1,933,300
RE103	98,600
RE104	2,860,900
RE105	402,800
RE106	78,600

RE107	86,800
RE108	144,600
RE109	122,100
STRP-101	168,900
STRP-102	205,400
WT-101	345,400
WT-102	134,500

9.6.3 Economic Analysis Calculation

Depreciation 7	Depreciation 5	Taxable Income 7	Taxable Income 5	Tax 7	Tax 5	Cash Flow 7	Cash Flow 5	Discount Factor	Present Value of CF 7	Present Value of CF 5	NPV 7	NPV 5
0.00	0.00	-185.04	-185.04	0.00	0.00	-185.04	-185.04	0.96	-178.38	-178.38	-178.38	-178.38
0.00	0.00	-555.12	-555.12	0.00	0.00	-555.12	-555.12	0.93	-515.85	-515.85	-694.23	-694.23
96.70	135.34	-57.70	-96.34	0.00	0.00	39.01	39.01	0.90	34.94	34.94	-659.29	-659.29
165.73	216.55	-40.84	-91.66	-20.19	-33.72	145.09	158.61	0.86	125.29	136.96	-534.00	-522.32
118.36	129.93	6.54	-5.04	-14.29	-32.08	139.19	156.97	0.83	115.86	130.67	-418.14	-391.65
84.52	77.96	40.37	46.94	2.29	-1.76	122.61	126.66	0.80	98.38	101.63	-319.75	-290.02
60.43	77.96	64.46	46.94	14.13	16.43	110.76	108.47	0.77	85.68	83.90	-234.07	-206.12
60.36	38.98	64.53	85.91	22.56	16.43	102.33	108.47	0.75	76.31	80.88	-157.77	-125.24
60.43	0.00	64.46	124.89	22.59	30.07	102.31	94.82	0.72	73.54	68.16	-84.23	-57.08
30.18	0.00	94.71	124.89	22.56	43.71	102.33	81.18	0.69	70.91	56.25	-13.32	-0.82
0.00	0.00	124.89	124.89	33.15	43.71	91.74	81.18	0.67	61.28	54.23	47.96	53.40
0.00	0.00	-6.18	-6.18	43.71	43.71	-49.90	-49.90	0.64	-32.13	-32.13	15.83	21.27
0.00	0.00	-6.18	-6.18	-2.16	-2.16	-4.02	-4.02	0.62	-2.50	-2.50	13.34	18.78
0.00	0.00	-6.18	-6.18	-2.16	-2.16	-4.02	-4.02	0.60	-2.41	-2.41	10.93	16.37
0.00	0.00	-6.18	-6.18	-2.16	-2.16	-4.02	-4.02	0.58	-2.32	-2.32	8.61	14.05
0.00	0.00	-6.18	-6.18	-2.16	-2.16	-4.02	-4.02	0.56	-2.24	-2.24	6.38	11.82
0.00	0.00	-6.18	-6.18	-2.16	-2.16	-4.02	-4.02	0.54	-2.15	-2.15	4.22	9.66
0.00	0.00	-6.18	-6.18	-2.16	-2.16	-4.02	-4.02	0.52	-2.08	-2.08	2.15	7.59
0.00	0.00	-6.18	-6.18	-2.16	-2.16	-4.02	-4.02	0.50	-2.00	-2.00	0.14	5.58
0.00	0.00	-6.18	-6.18	-2.16	-2.16	-4.02	-4.02	0.48	-1.93	-1.93	-1.79	3.65
0.00	0.00	-6.18	-6.18	-2.16	-2.16	-4.02	-4.02	0.46	-1.86	-1.86	-3.65	1.79
0.00	0.00	-6.18	-6.18	-2.16	-2.16	-4.02	-4.02	0.45	-1.79	-1.79	-5.44	0.00

Figure 15: Excel of Sensitivity Analysis Calculation

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